

Language Priors for Visual Intelligence

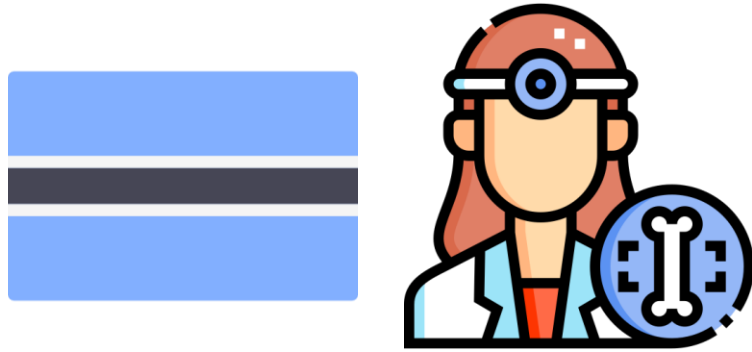
Dissertation Defense

Yue Yang

Advisor: Mark Yatskar

Committee: Dan Roth (Chair), Chris Callison-Burch,
Dinesh Jayaraman, Pratik Chaudhari, Jacob Andreas

Build Visual Intelligence to Benefit the Real World



Only **15** Radiologists
in Botswana



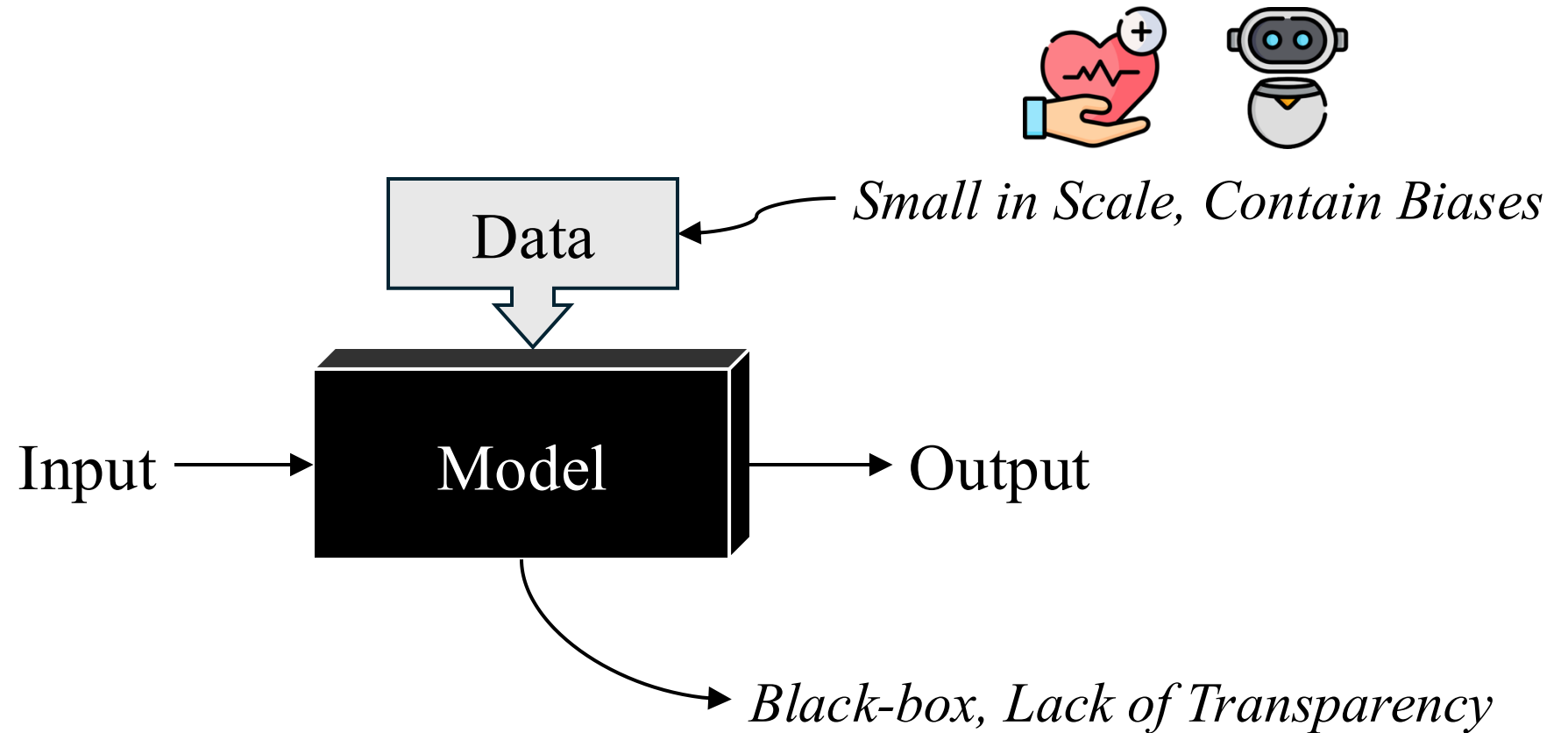
2.6M people



2000 Patient Died
for TB Each year

Are AI systems ready to be deployed in critical domains?

Challenges of Existing Methods in Critical Domains



Limited Generalizability in Critical Domains.

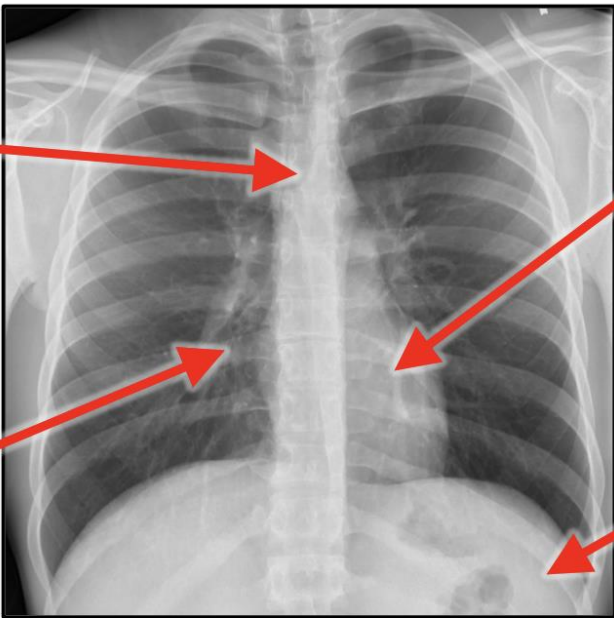
Catastrophic Failure in Critical Domains



Leverage Knowledge Priors

AIRWAY
Trachea
Bronchi

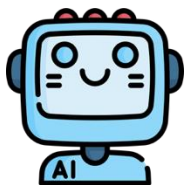
BREATHING
Lung
Pleura



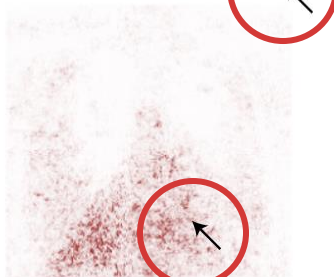
CARDIAC
Heart

DIAPHRAGM
Costophrenic
angles

EXTRA
Pacemaker Tubes

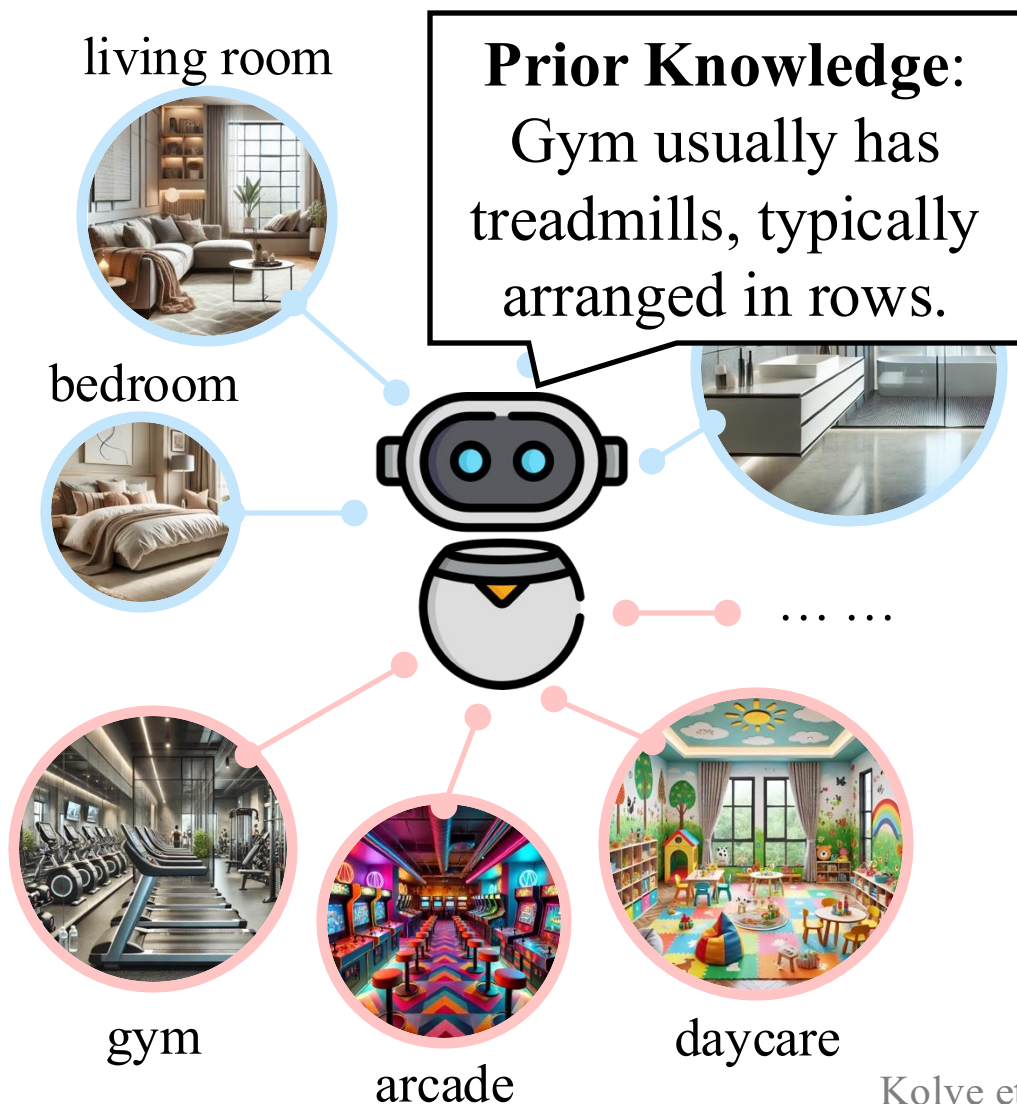


Take Shortcuts



saliency maps

Limited Generalization in Unseen Scenarios

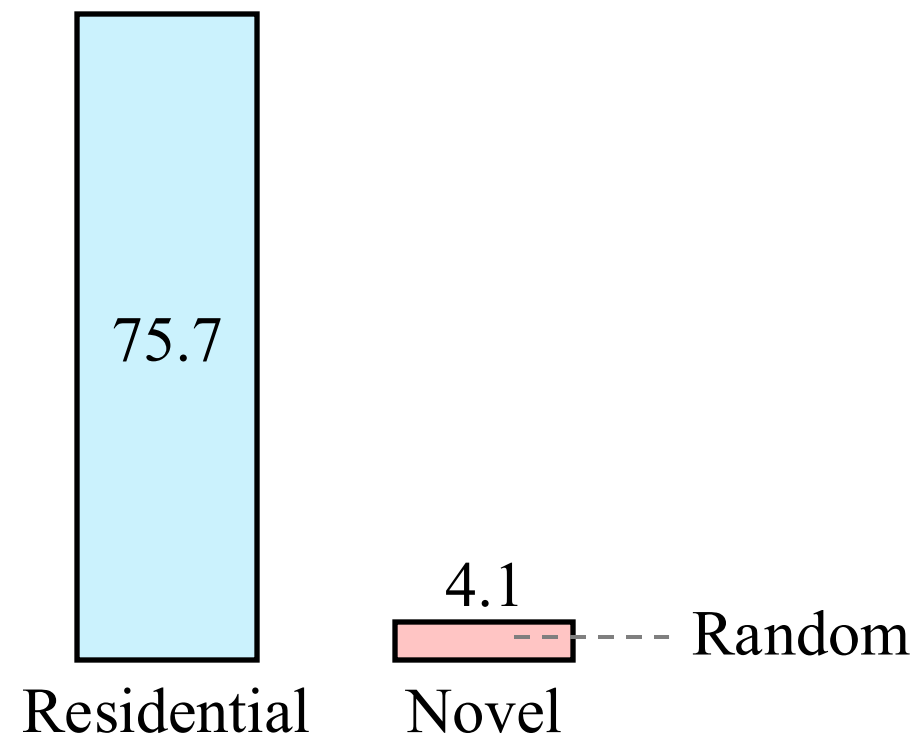


Residential Scenes
ProcTHOR, iTHOR

Novel Scenes
Gym, Arcade, etc.

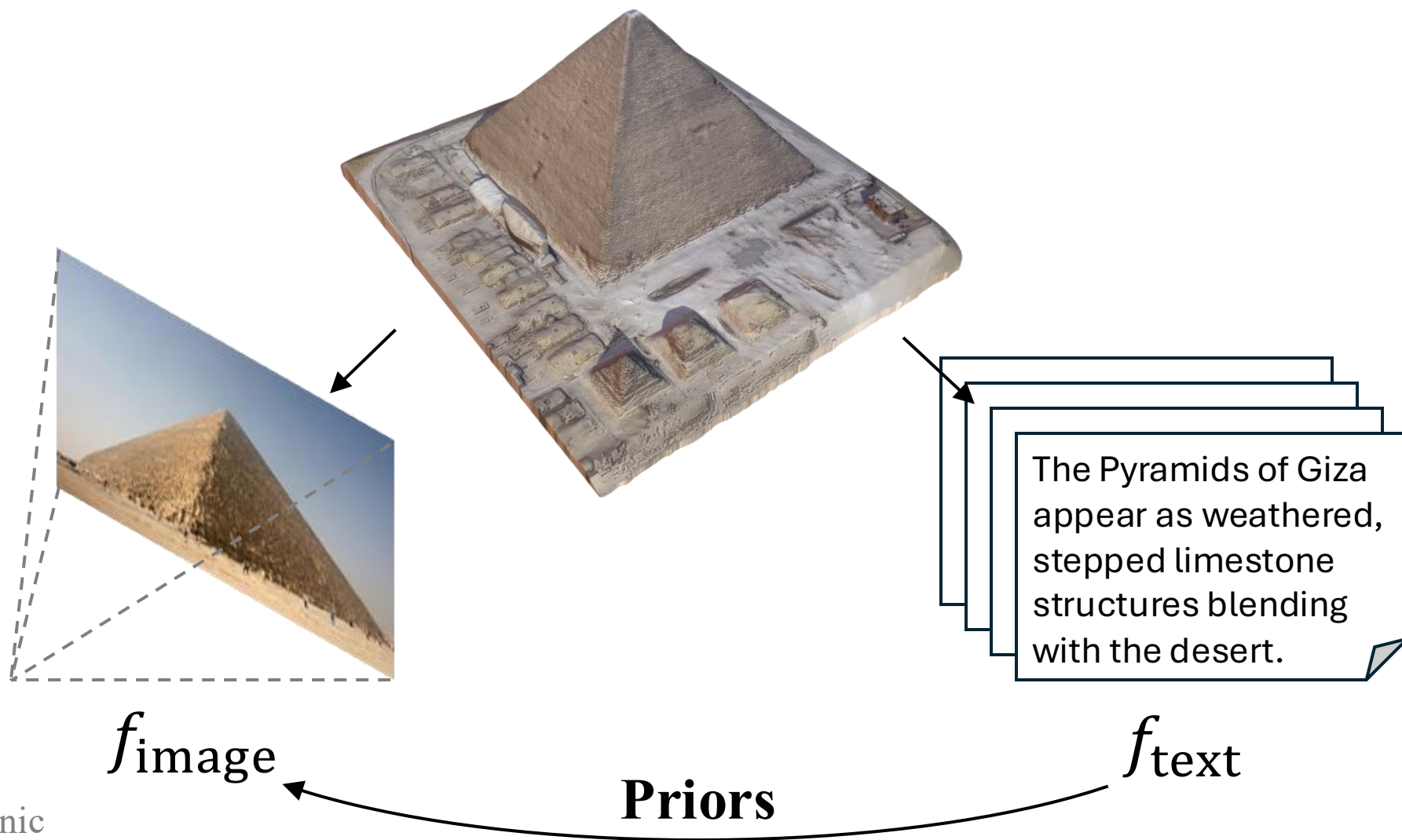
0-shot trained on ProcTHOR-10K

ObjectNav
Success Rate (%)

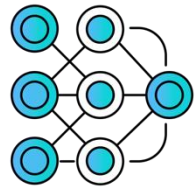


Kolve et al. Ai2-THOR. 2017
Deitke et al. ProcTHOR. 2022

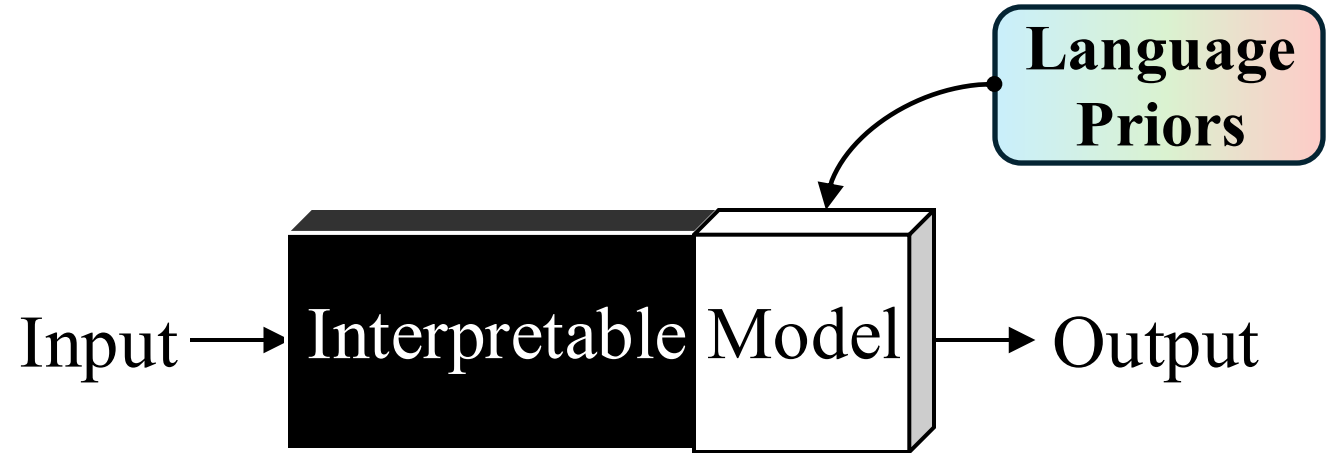
Language and Vision Represent the Same World



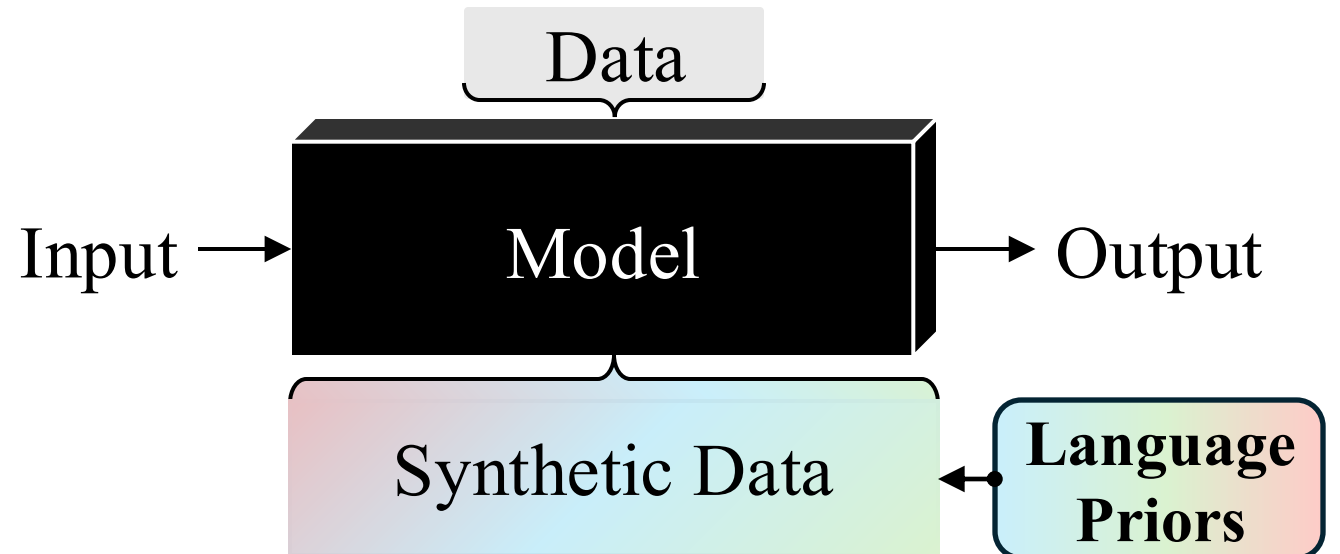
How to Apply Language Priors?



Language as
Model Structure



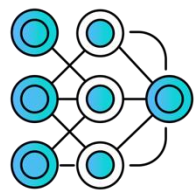
Language to Create
Synthetic Data



Thesis Statement

By integrating *language priors* into **model structure** and **data synthesis**, this thesis aims to enhance the **interpretability, robustness, and data efficiency** of *visual intelligence*.

Talk Outline

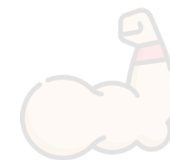


Language as
Model Structure



Interpretability

Language Bottleneck
(CVPR 23)



Robustness

Knowledge Bottleneck
(NeurIPS 24, Spotlight)



Data Efficiency



Holodeck
(CVPR 24)

Embodied AI

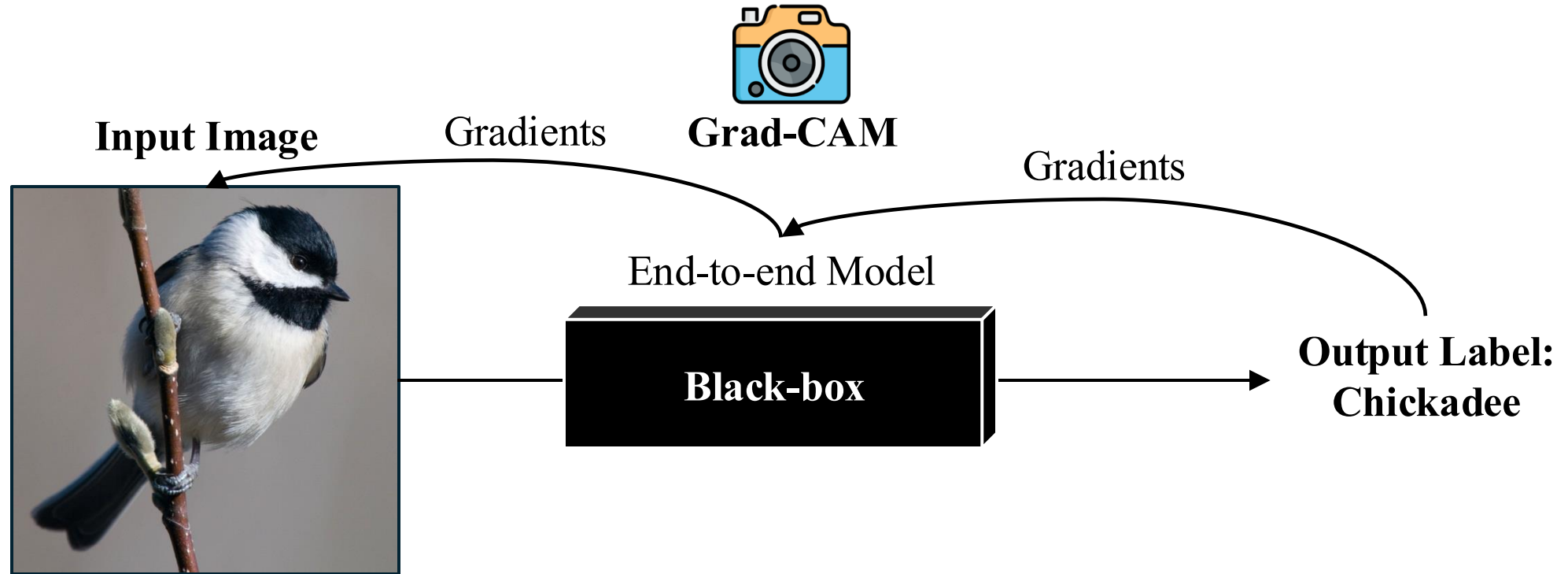
Code Guided Synthetic
Text-rich Data

VLM



Language to Create
Synthetic Data

Post-hoc Interpretable Method



[1] Selvaraju et al. Grad-CAM. ICCV 2017.

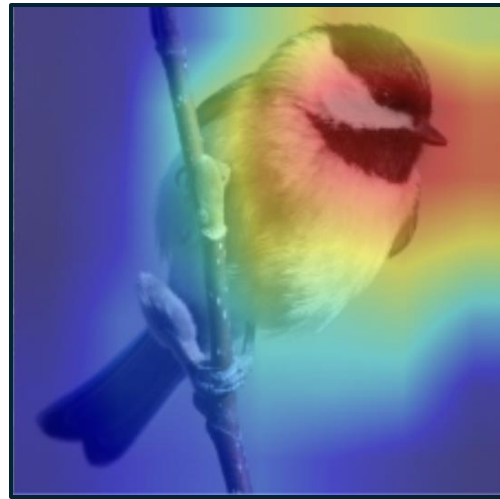
[2] Cynthia Rudin. Stop explaining black box machine learning models for high stakes decisions and use interpretable models instead. Nature Machine Intelligence. 2019.

Post-hoc Interpretable Method

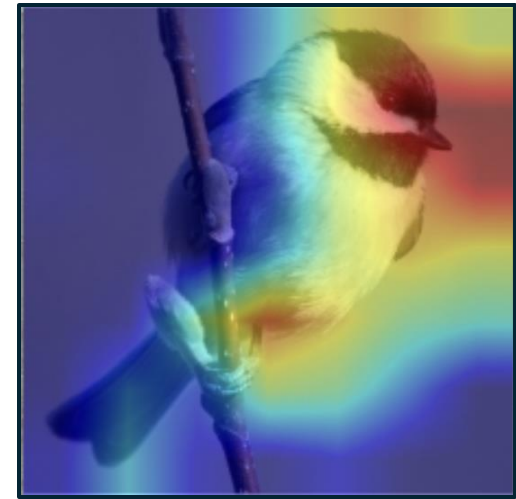
Input Image



Target Class =
Chickadee



Target Class =
Coffee Maker

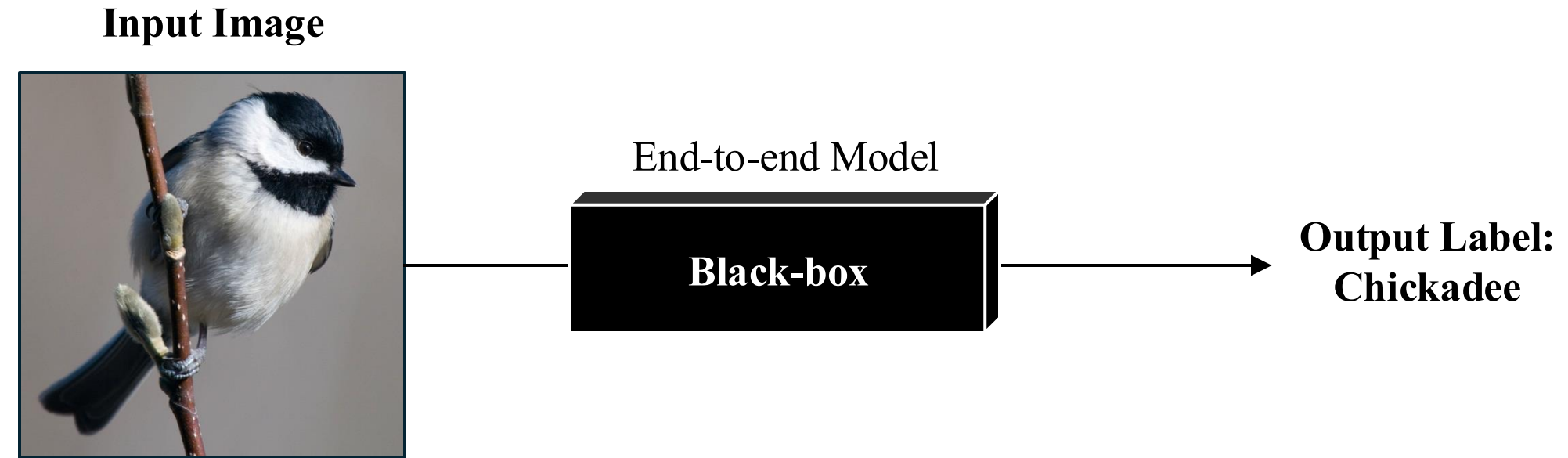


Explanations are **not faithful** and **misleading**.

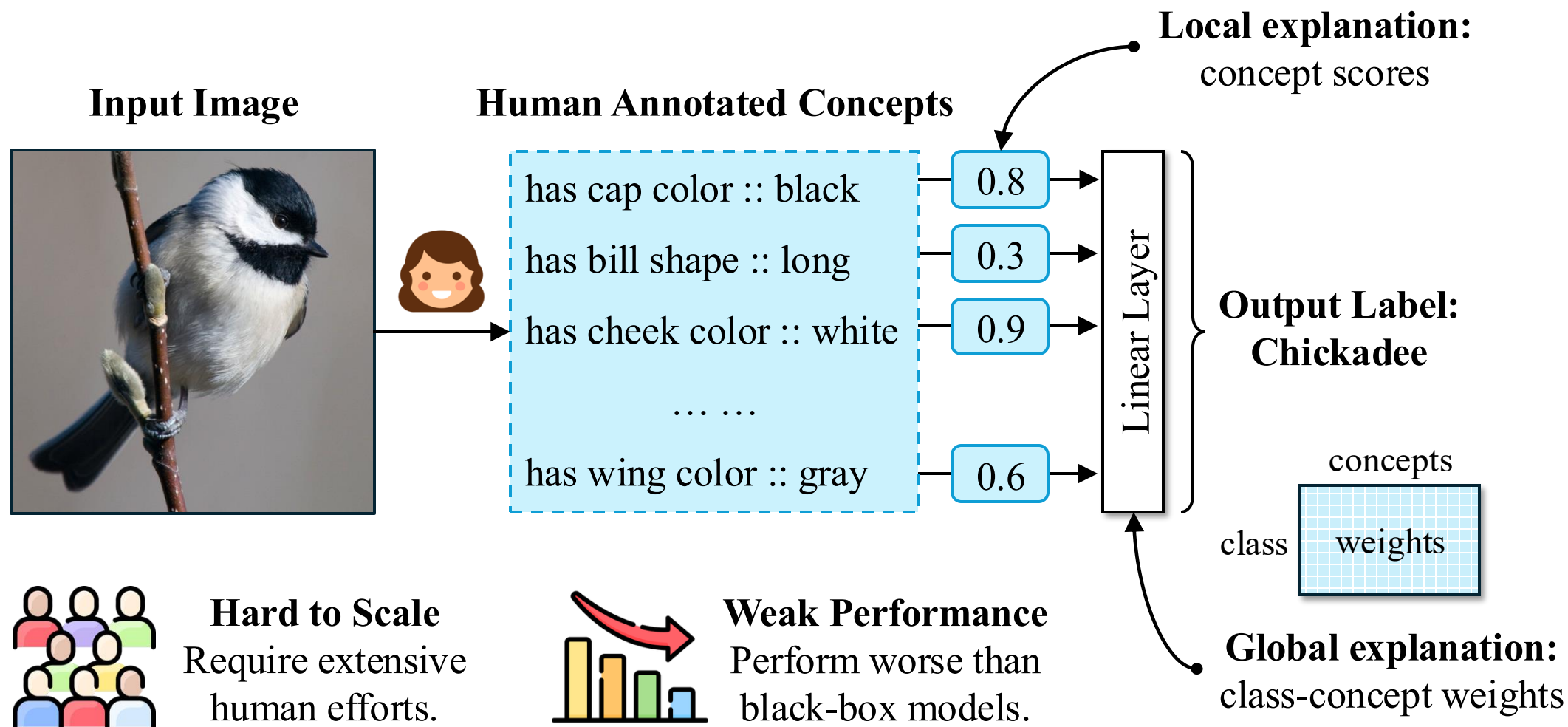
[1] Selvaraju et al. Grad-CAM. ICCV 2017.

[2] Cynthia Rudin. Stop explaining black box machine learning models for high stakes decisions and use interpretable models instead. Nature Machine Intelligence. 2019.

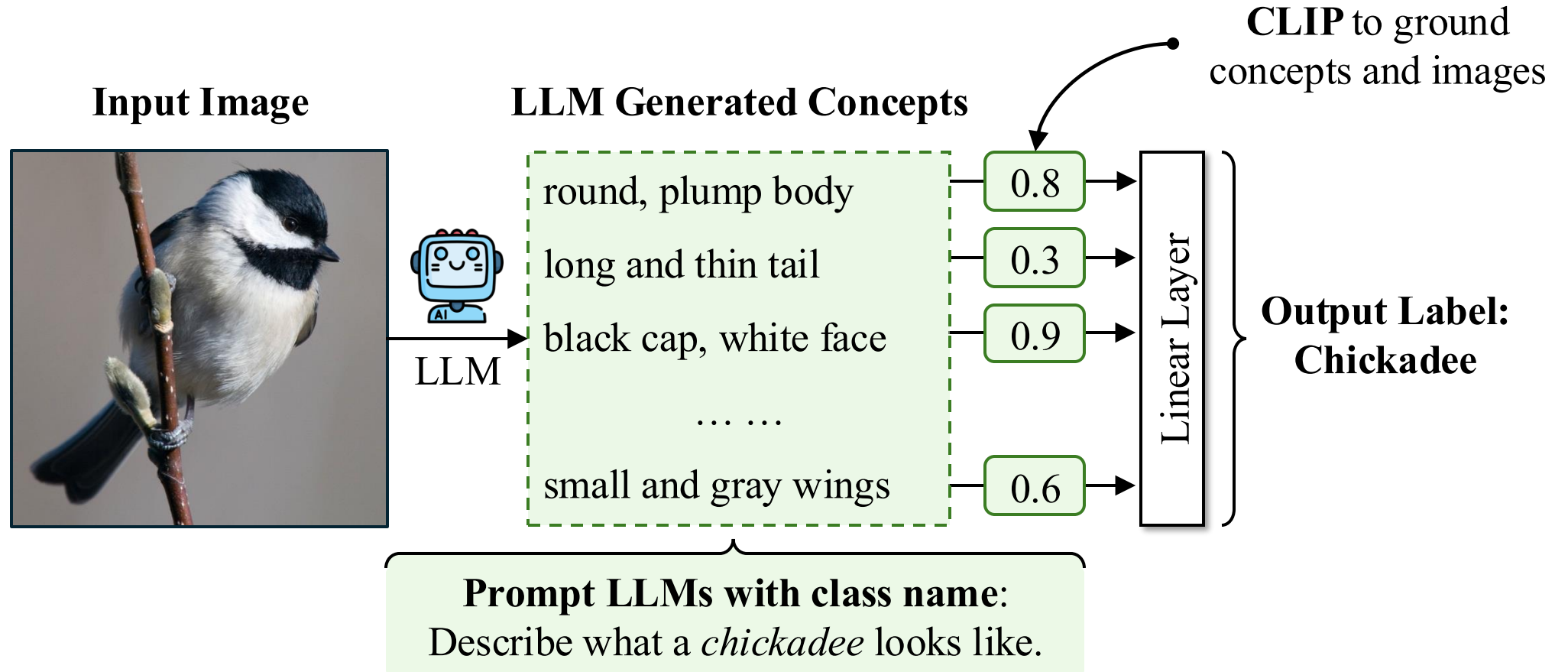
Inherently Interpretable: Concept Bottleneck Model



Inherently Interpretable: Concept Bottleneck Model



Language Model Guided Bottleneck (LaBo)



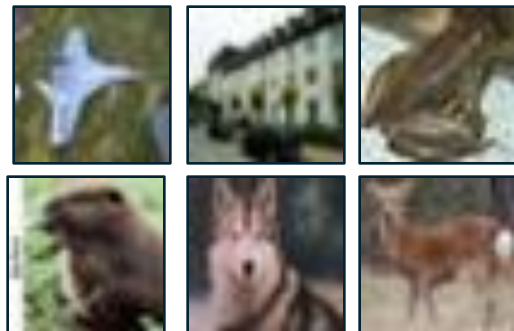
Evaluate LaBo on **Diverse** Classification Datasets

Common Objects

ImageNet1K



CIFAR-10/CIFAR-100

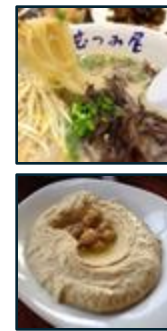


Fine-grained Objects

Flower-102



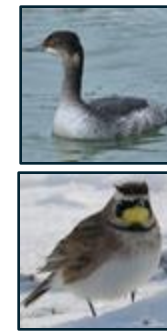
Food-101



Aircraft

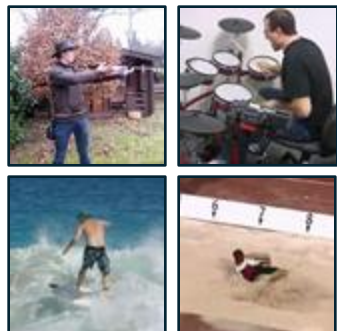


CUB



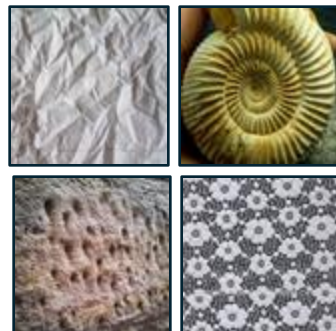
Action

UCF-101



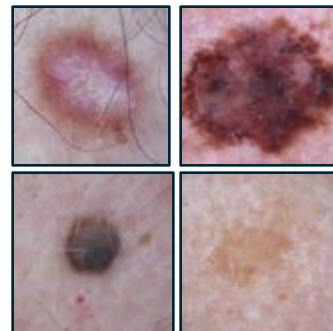
Textures

DTD



Skin Tumors

HAM10000



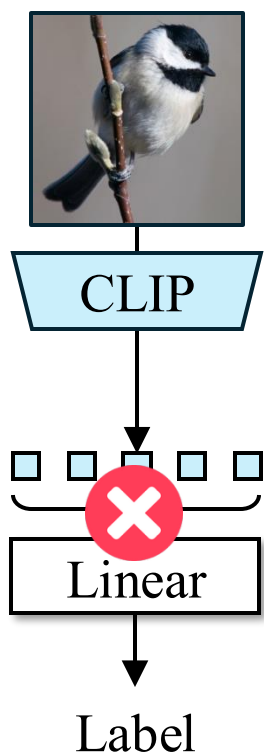
Satellite

RESISC45



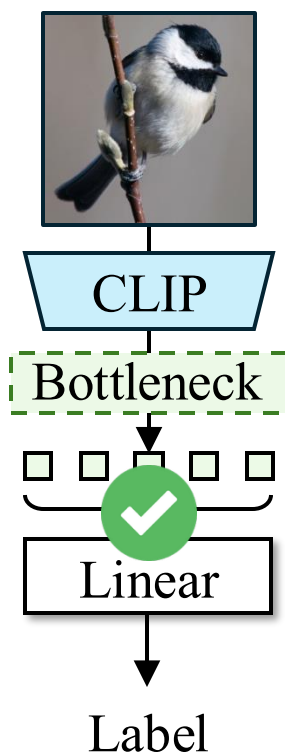
LaBo is Competitive with Black-box Model

Linear Probe

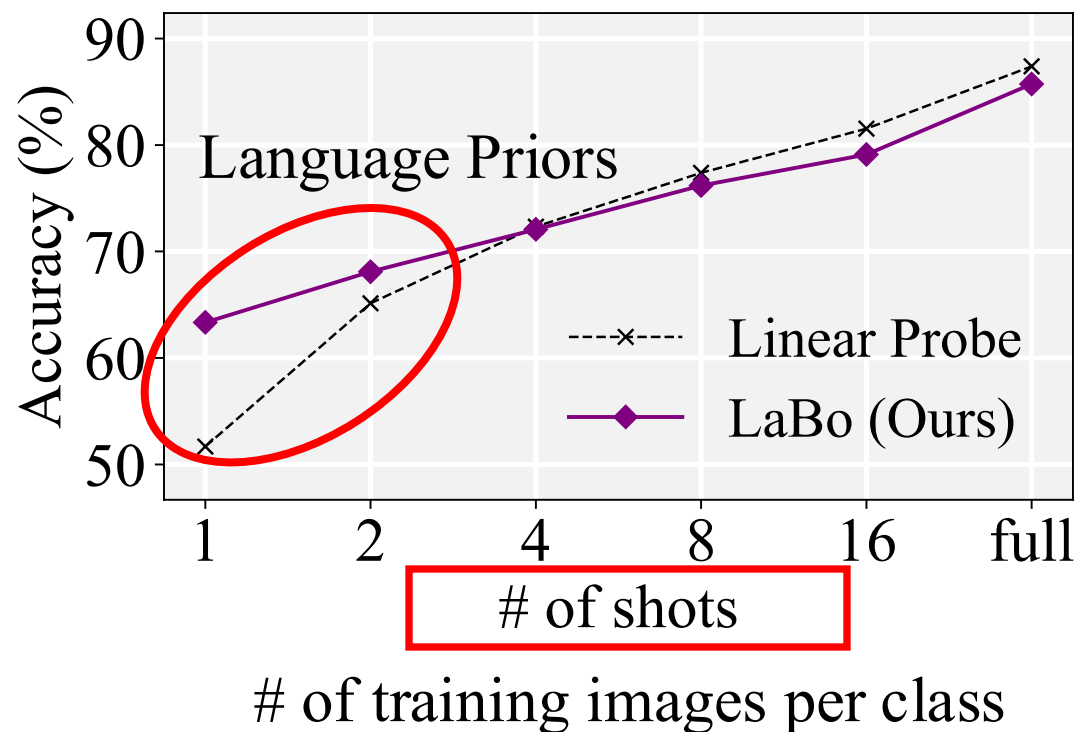


Interpretable?

LaBo (Ours)



Few-shot / Fully Supervised Performance



Qualitative Examples to Show Interpretability

Ranked by the weights in the linear layer.

class

concepts

weights

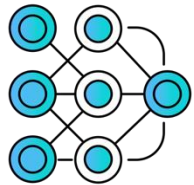
	Class Name	Top-3 Concepts	Class Name	Top-3 Concepts
ImageNet	badger 	1. short legs and long body make it an excellent digger 2. black-and-white striped fur 3. coat is very shaggy	ant 	1. black and red stinger 2. small, black insect with six legs 3. long, slender antennae that it uses to smell and touch

LLMs can construct **high-performance** concept bottlenecks for **interpretable** image classifiers.

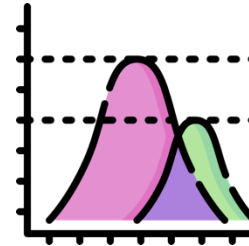
Apply Bottlenecks in Medical Domains



Data in small-scale,
and has biases.



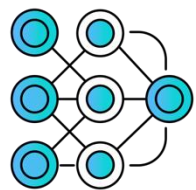
Models lack priors for
processing medical images.



Sensitive to
Domain Shifts

We need more **robust** models!

Talk Outline

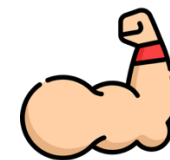


Language as
Model Structure



Interpretability

Language Bottleneck
(CVPR 23)



Robustness

Knowledge Bottleneck
(NeurIPS 24, Spotlight)



Data Efficiency



Language to Create
Synthetic Data



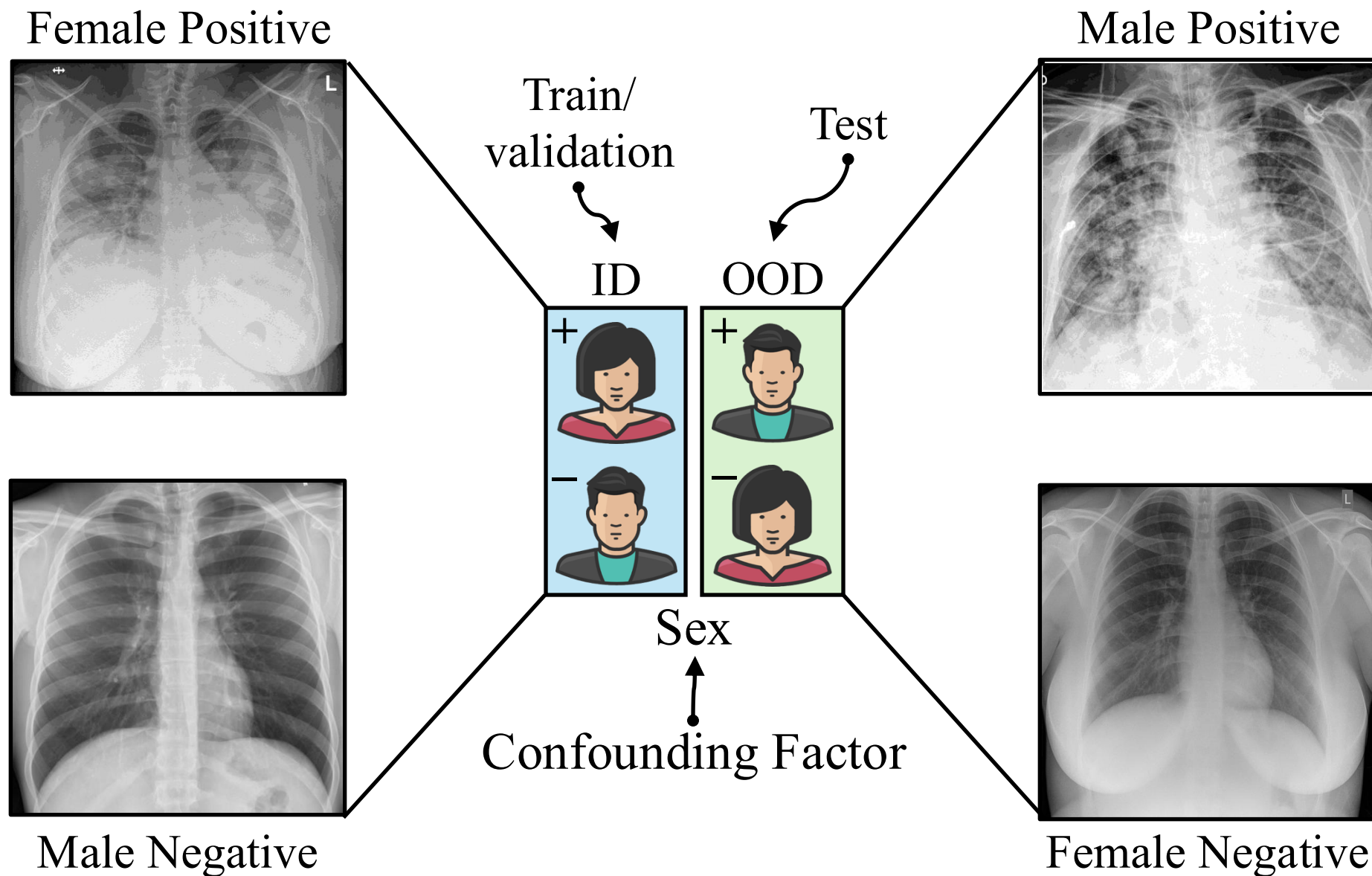
Holodeck
(CVPR 24)

Embodied AI

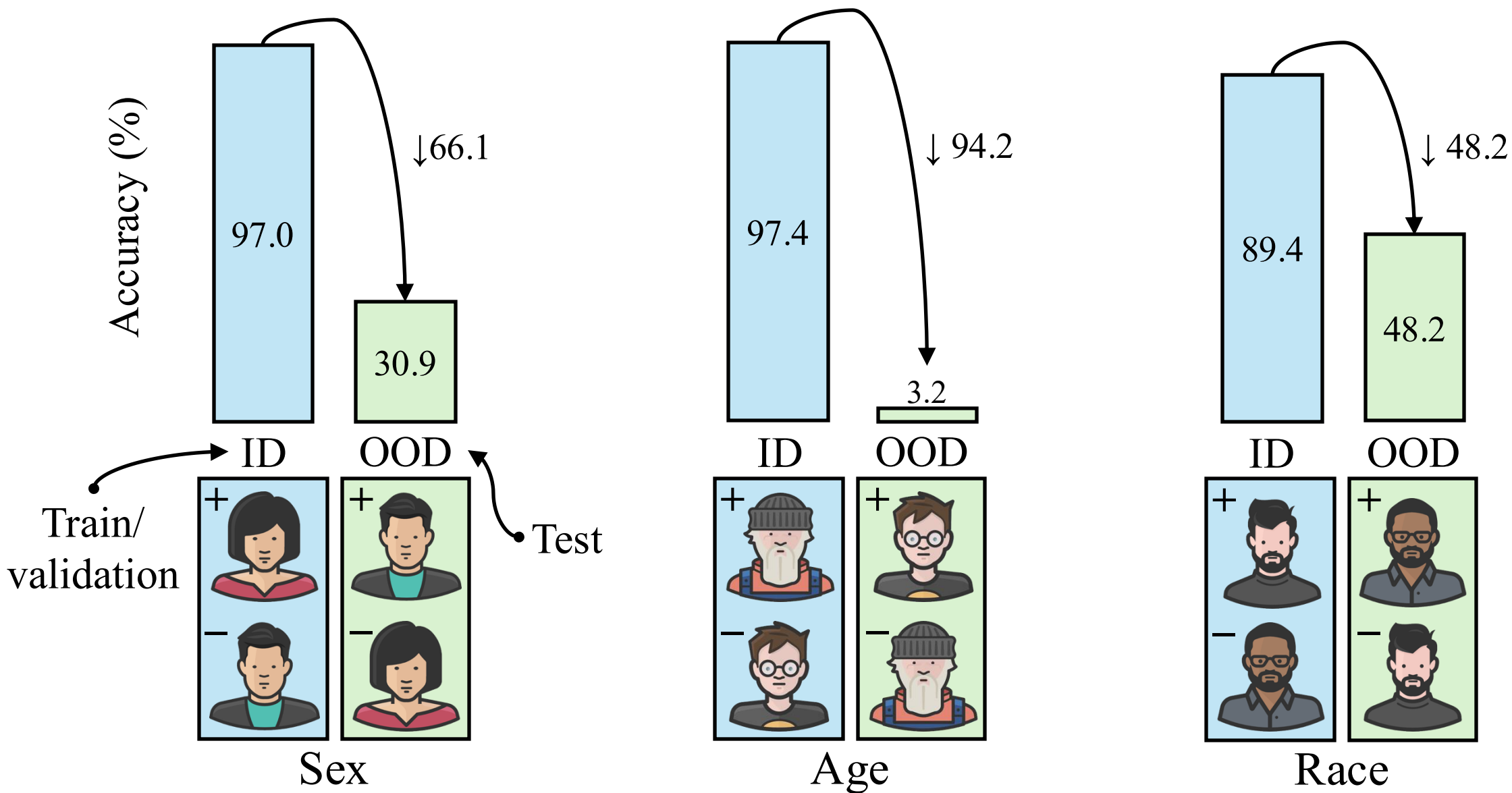
Code Guided Synthetic
Text-rich Data

VLM

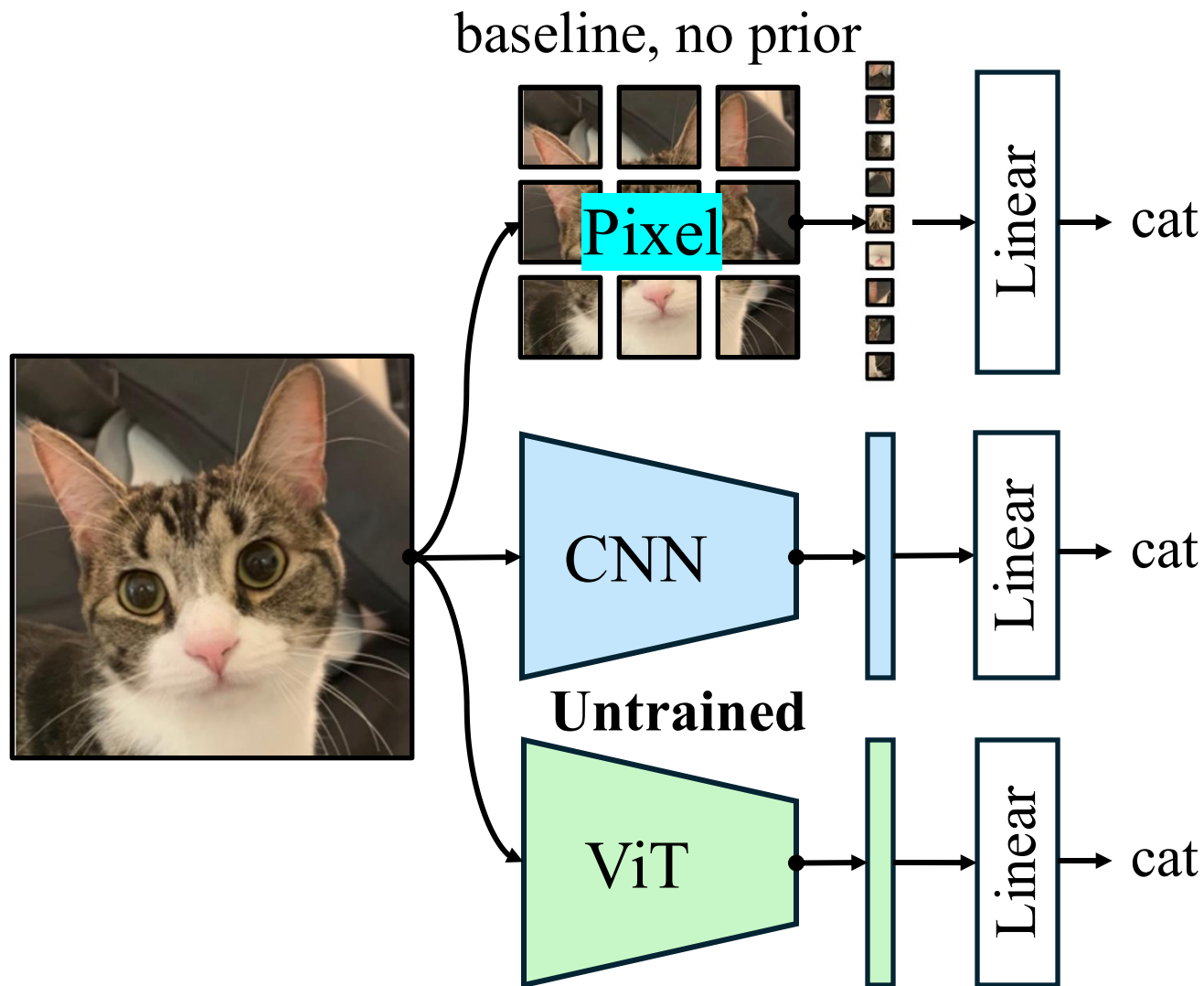
Confounded Dataset to Test Robustness



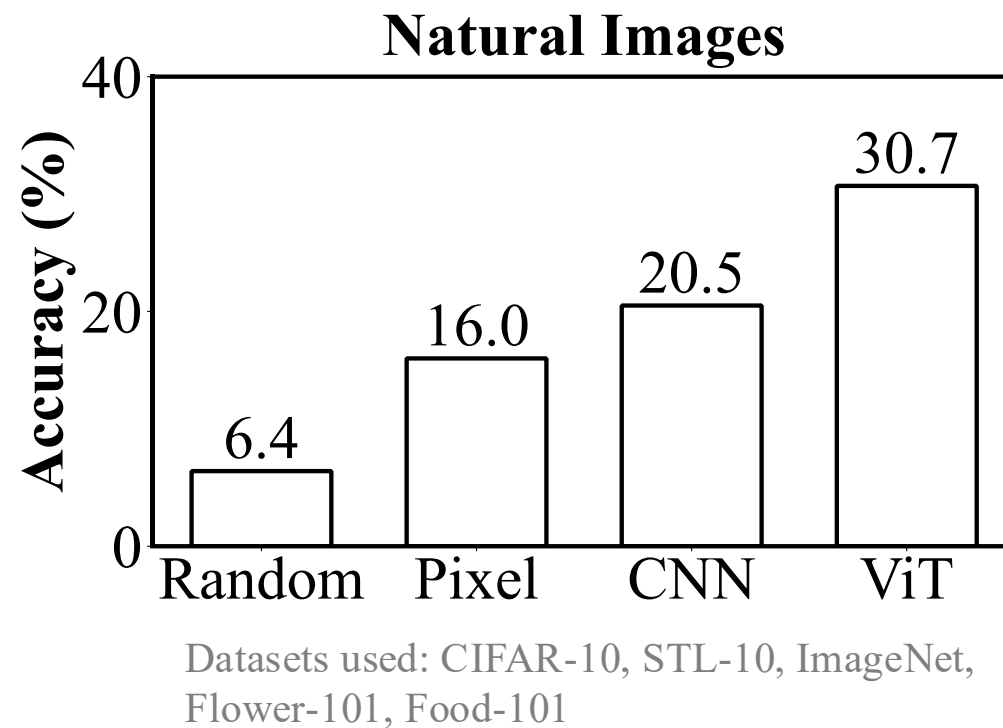
Black-box is Sensitive to Domain Shifts



Deep Image Priors



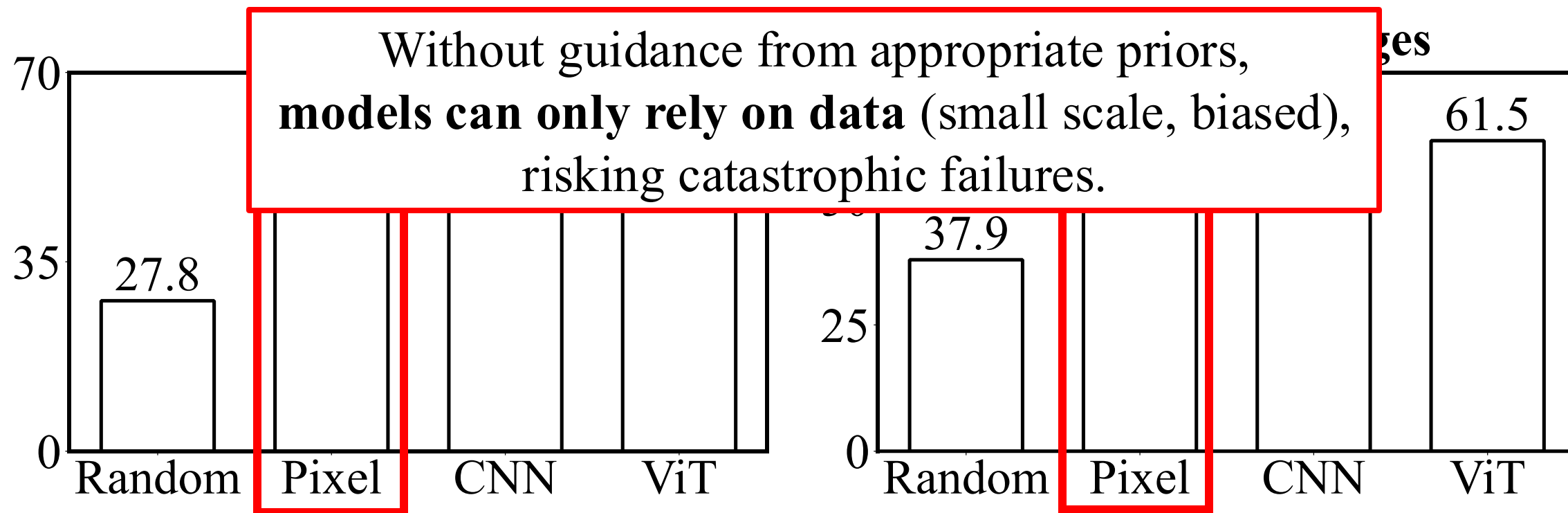
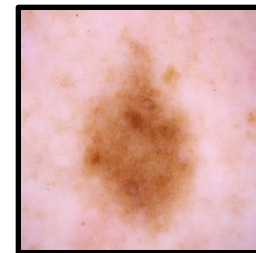
Vision backbones have a **deep image prior**, even when entirely untrained.



A. M. Saxe et al., On random weights and unsupervised feature learning. ICML 2011.

D. Ulyanov et al., Deep image prior. CVPR 2018.

Lack of Priors for Medical Images



X-ray Datasets: Pneumonia, COVID-QU, NIH-CXR, Open-I, VinDr-CXR.

Skin Lesion Datasets: HAM10000, BCN20000, PAD-UFS-20, Melanoma, UWaterloo.

How to Incorporate More Priors?

Inspired by medical education.

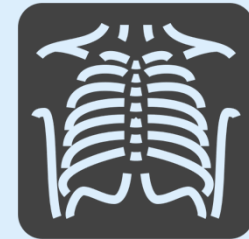
Medical School



Textbooks

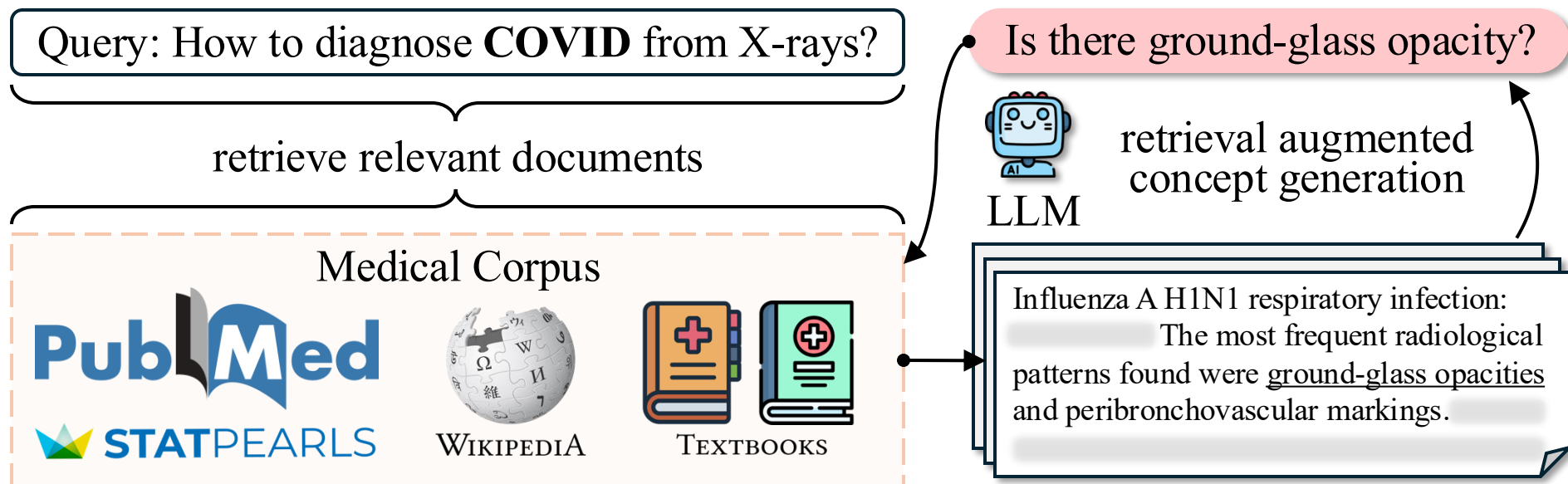


Residency in Hospital



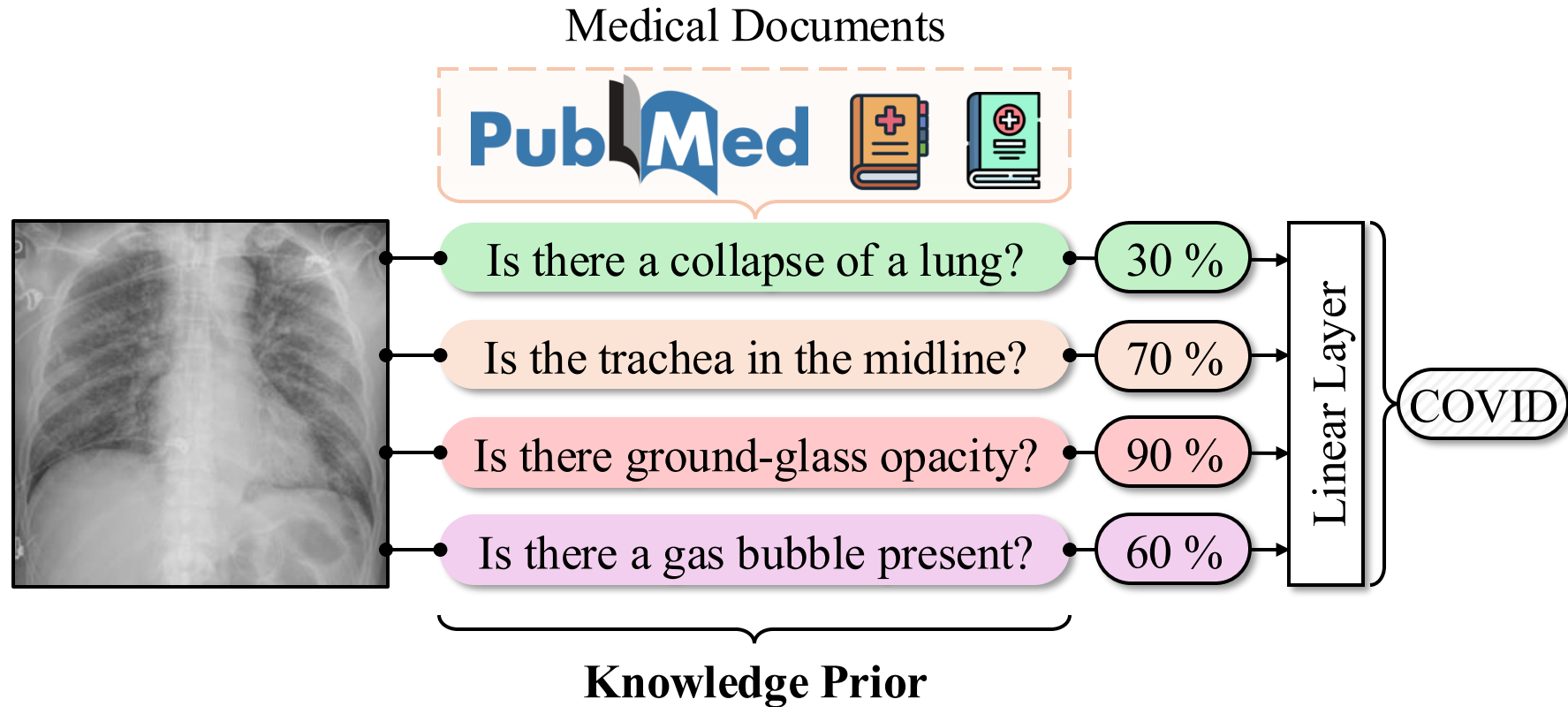
Practice

Knowledge Prior from Medical Documents



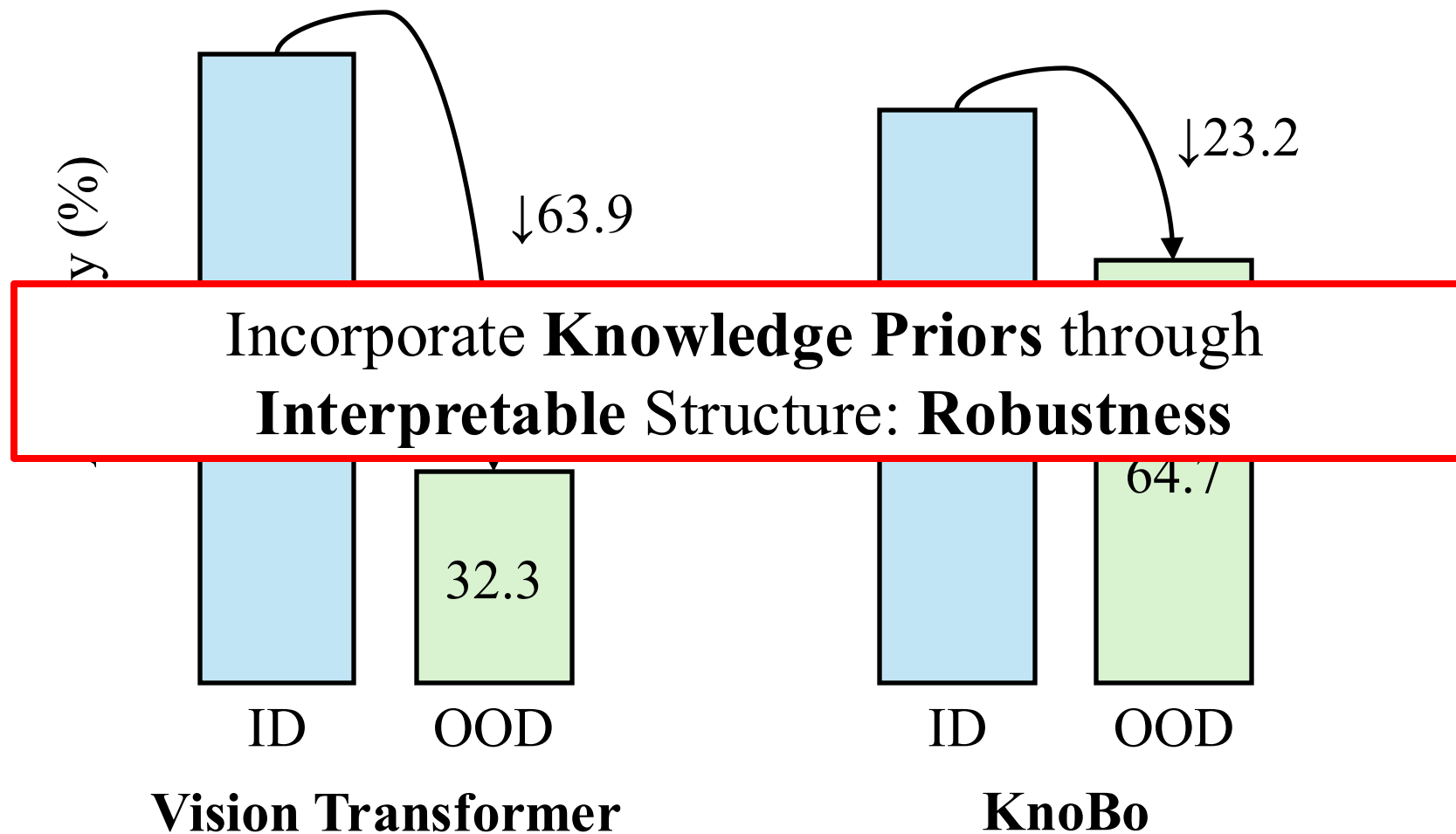
PubMed 5M Articles, 300M+ paragraphs

Knowledge-enhanced Concept Bottlenecks (KnoBo)



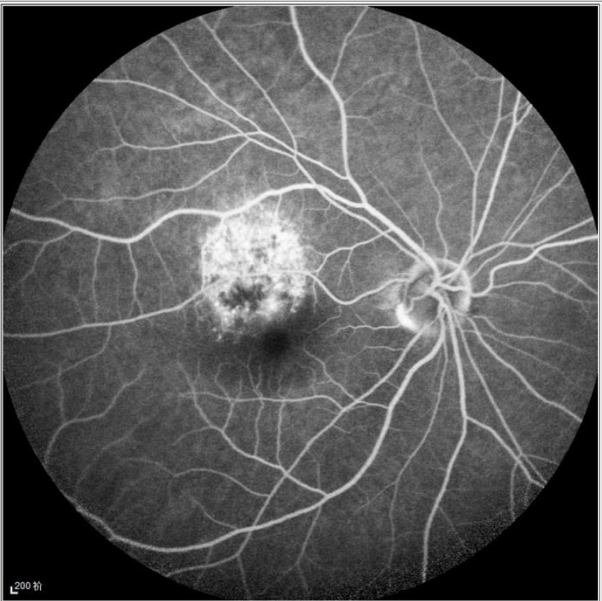
KnoBo is **more Robust** to Domain Shifts

Average Performance on 10 Confounded Datasets



Concept Bottlenecks can Assist Doctors

Choroid Melanoma

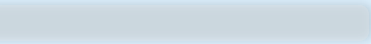


Rare: 5.1 per million
Metastatic Mortality:
4-12 months

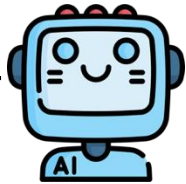


Analyze this case.

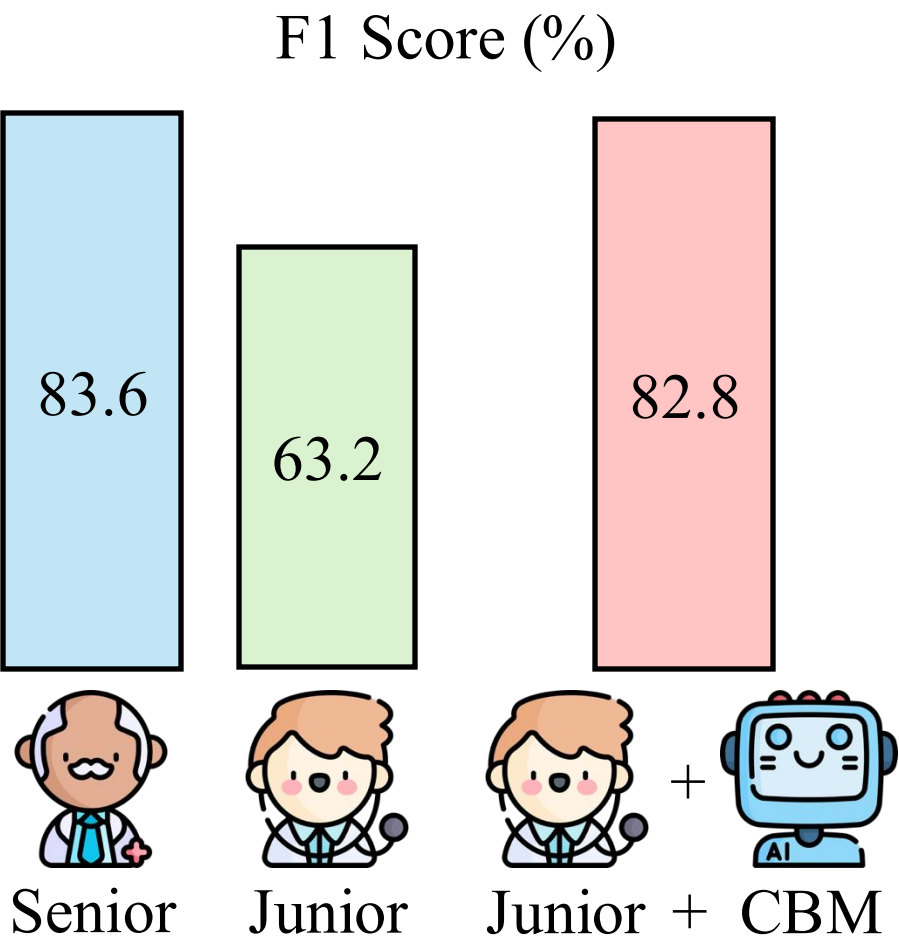
Top Concepts:

- 1. abnormal vein
- 2. pigment change
- 3. 
-

Final Prediction:
Malignant

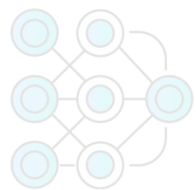


CBM



Doctors are from Beijing Tongren Hospital.

Talk Outline

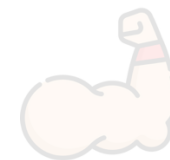


Language as
Model Structure



Interpretability

Language Bottleneck
(CVPR 23)



Robustness

Knowledge Bottleneck
(NeurIPS 24, Spotlight)



Data Efficiency



Language to Create
Synthetic Data



Holodeck
(CVPR 24)

Embodied AI

Code Guided Synthetic
Text-rich Data

VLM

Data Scarcity of Vision Tasks

130K episodes
17 month to collect

RT-1

ImageNet

LAION-400M
Image-text

Alt Text:
now at victorian
plumbing.co.uk

FLAN
1.6T tokens

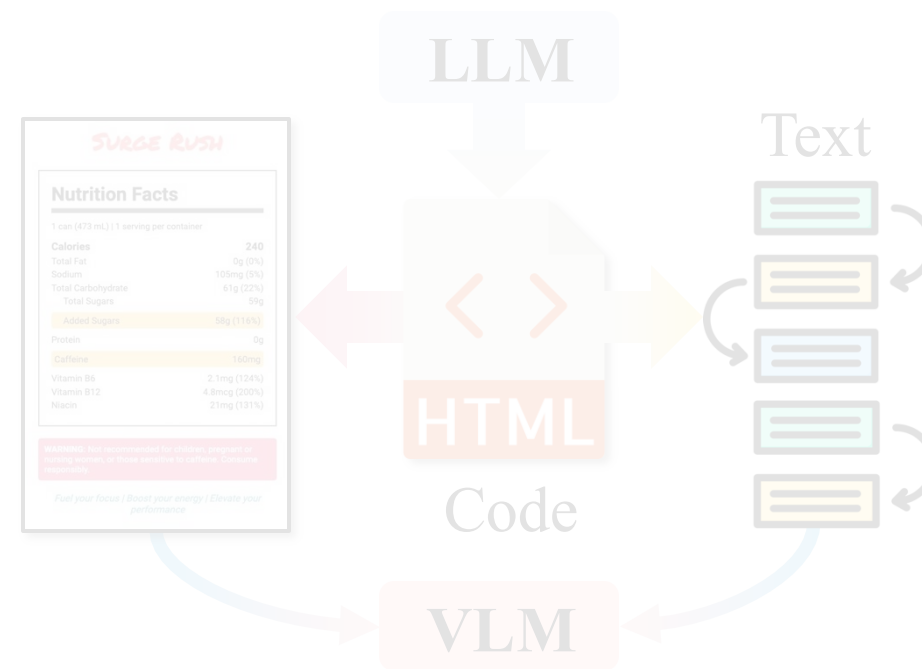
Brohan et al. RT-1. 2022. Deng et al. ImageNet. 2009.
Schuhmann et al. LAION-400M. 2021. Wei et al. FLAN. 2021.



Language Guided Synthetic Data



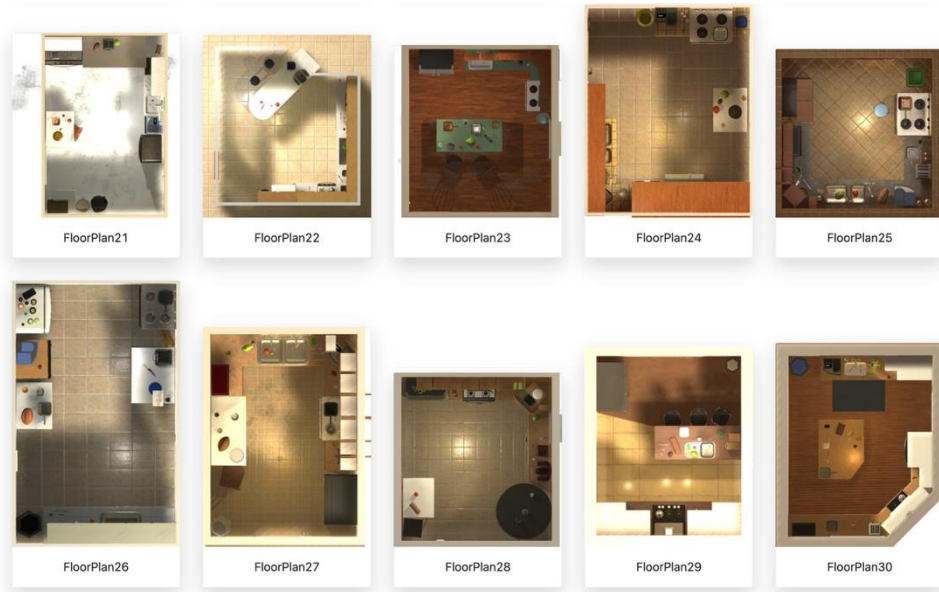
Language Guided
3D Environments Generation



Code Guided
Synthetic Text-rich Images

Limited Scale and Diversity of Environments

Previous works manually designed scenes.



iTHOR (120 Scenes, 4 types)

Kolve et al. 2017.



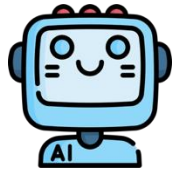
BEHAVIOR-1K (50 Scenes, 8 types)

Li et al. CoRL. 2022.

Motivation

LLMs contain the common-sense knowledge about designing environments.

How to design an office?

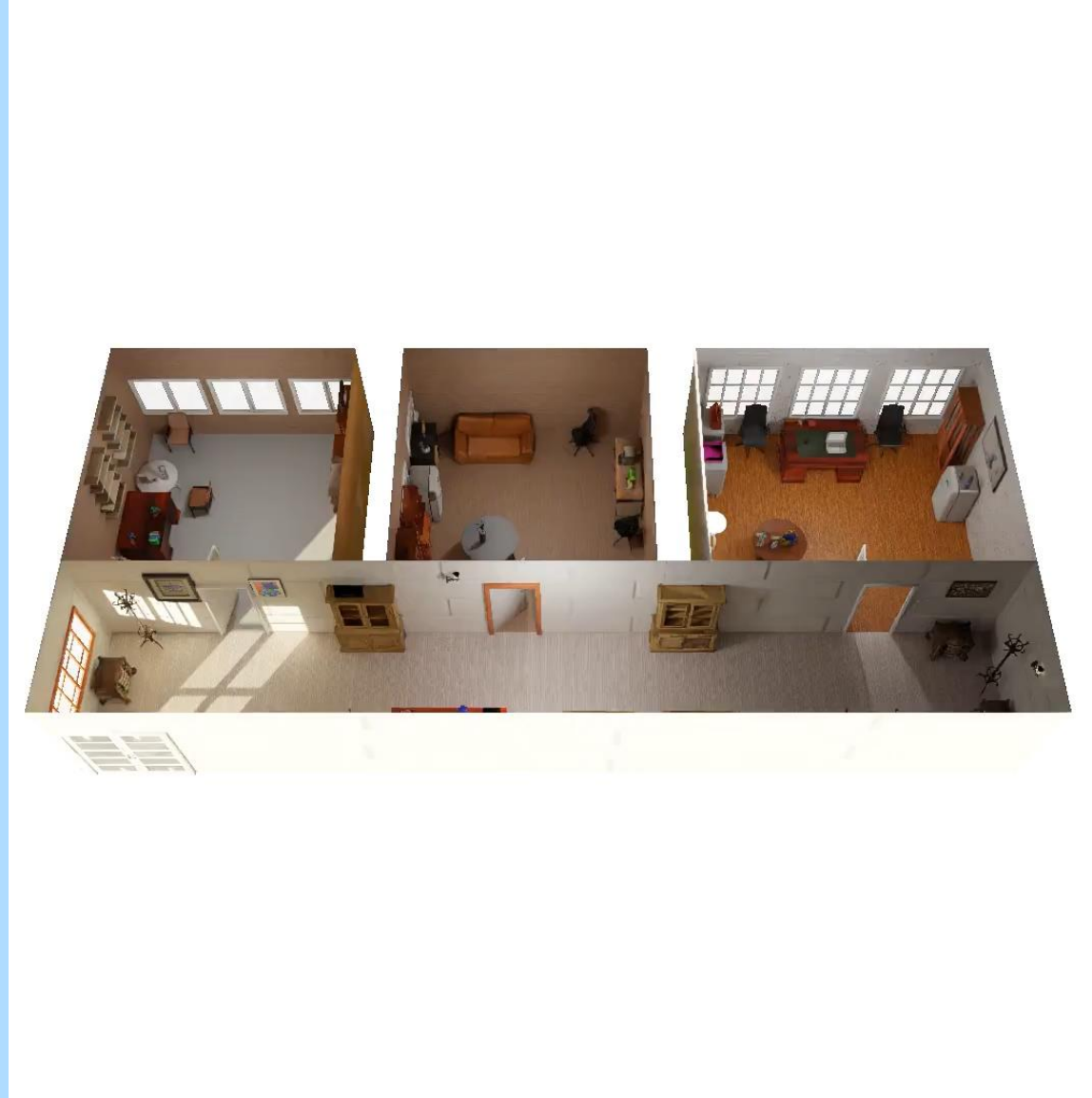


LLM

Place a desk and ergonomic chair at the center or near a window, with storage units like shelves or drawers nearby for easy access. Add a lamp, a few plants, and personal decor items on the desk or shelves, keeping the workspace tidy and organized.

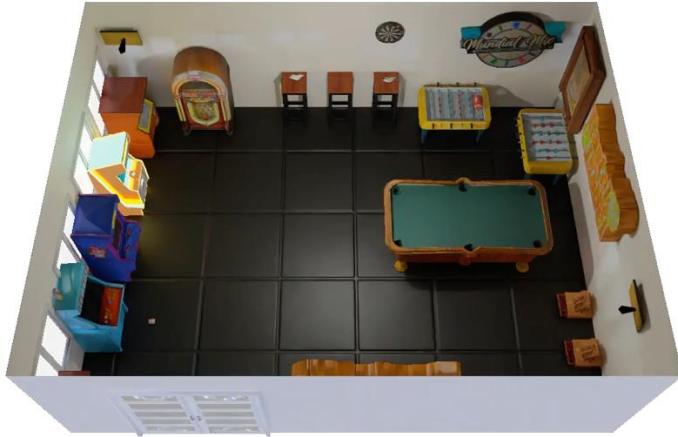


Three professors' office connected to a long hallway, the professor in office-1 is a fan of Star Wars.

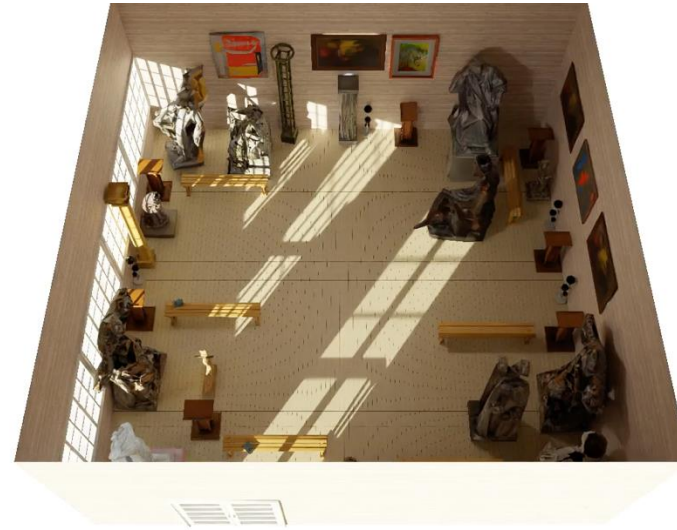




arcade room



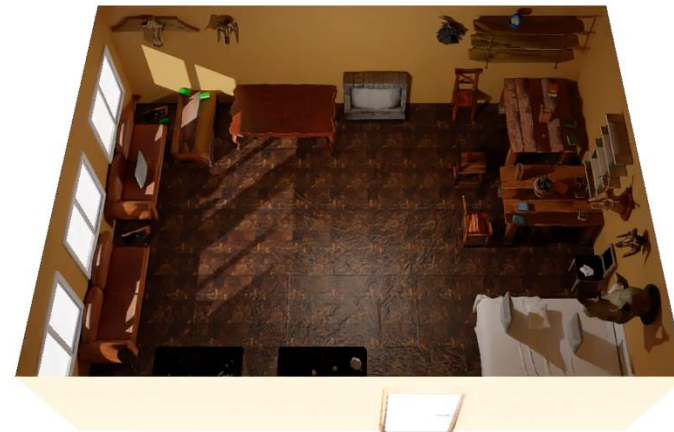
sculpture museum



spa



hunter cabin



modern style



Victorian style



Bohemian style



Industrial style



Generalize to Novel Environments

Zero-shot Object Navigation in Novel Environments



Novel object: piano



Novel Environment: music room



Synthesize

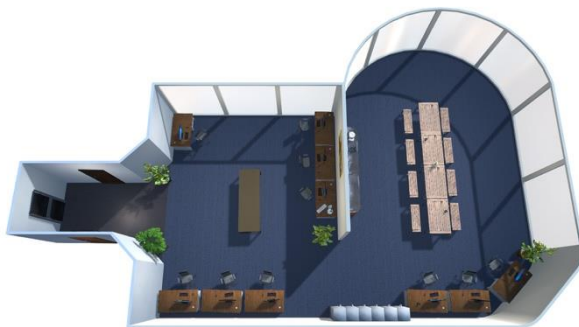


Generalize to Novel Environments

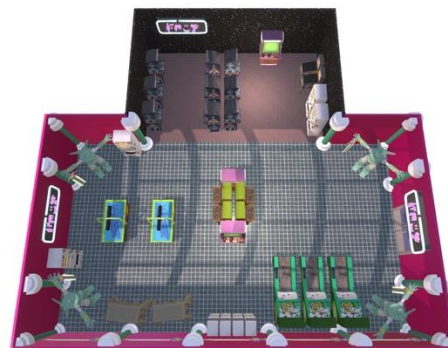
Daycare



Office



Arcade



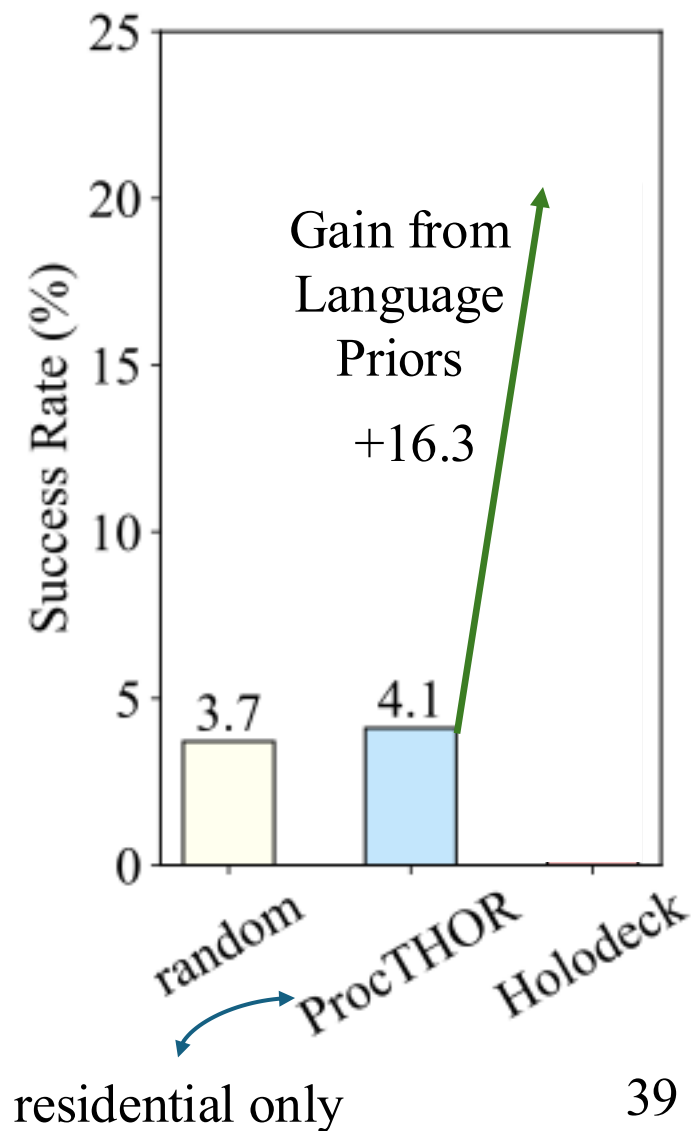
Gym



Music Room



10 Novel
Environments
Built by Experts
for Testing



Talk Outline

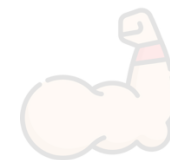


Language as
Model Structure



Interpretability

Language Bottleneck
(CVPR 23)



Robustness

Knowledge Bottleneck
(NeurIPS 24, Spotlight)



Data Efficiency



Language to Create
Synthetic Data



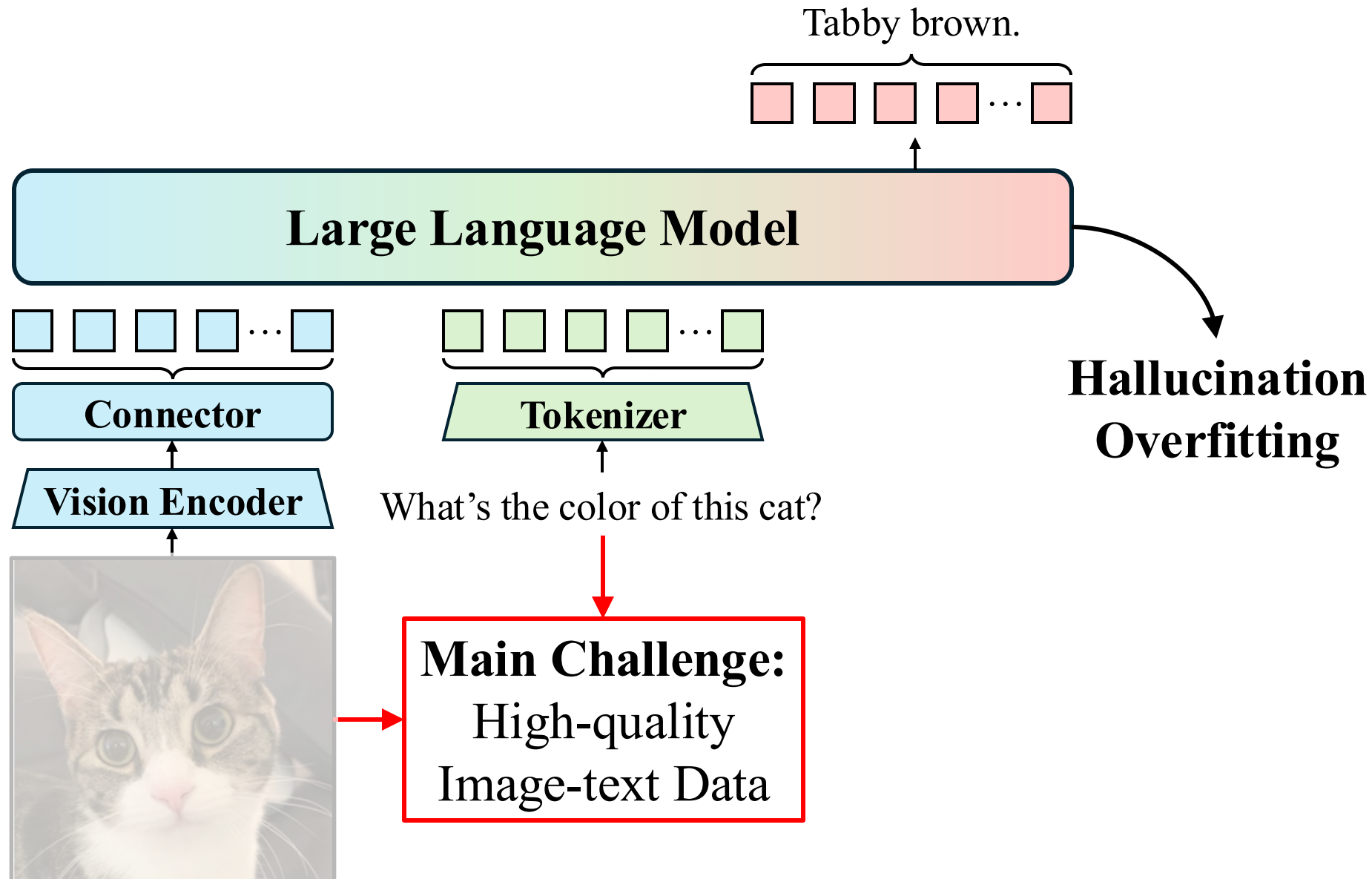
Holodeck
(CVPR 24)

Embodied AI

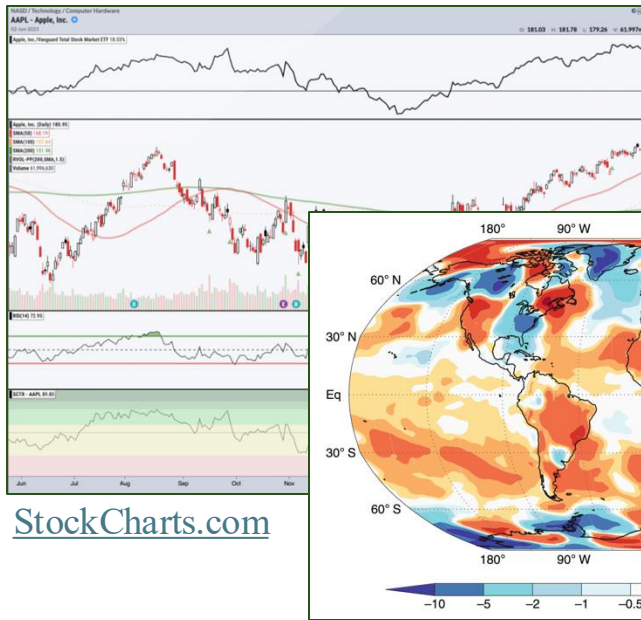
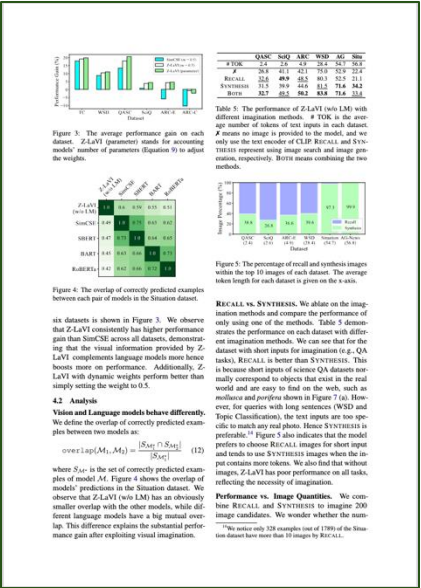
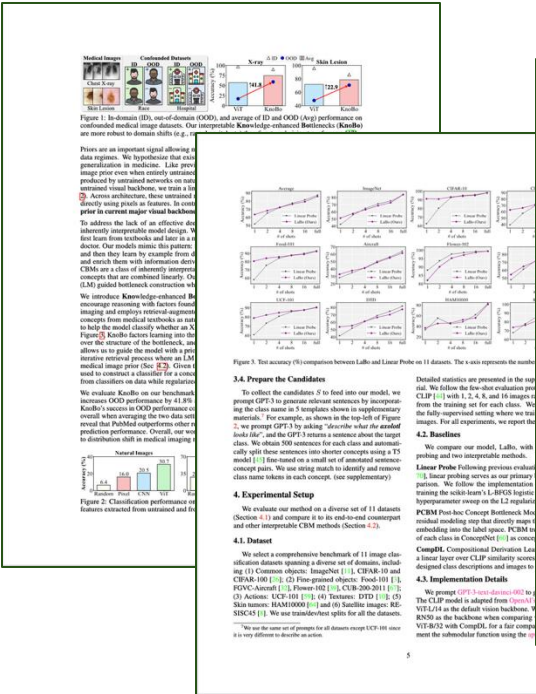
Code Guided Synthetic
Text-rich Data

VLM

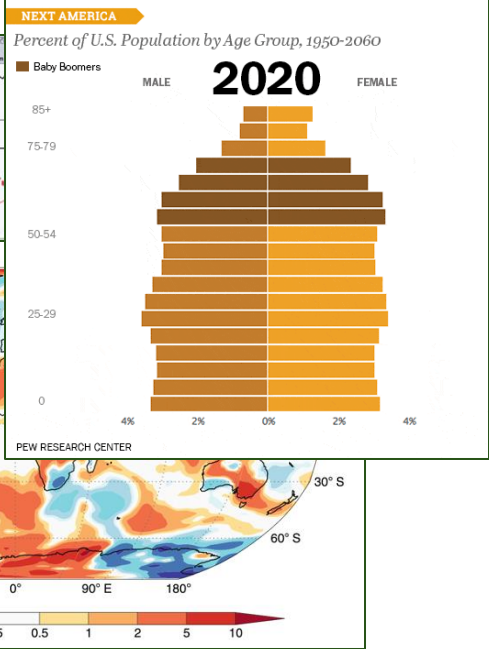
Vision Language Model



Text and Figure-rich Images



<https://www.pewresearch.org/>



(Seung-Ki Min, 2020)



Analyze Literature



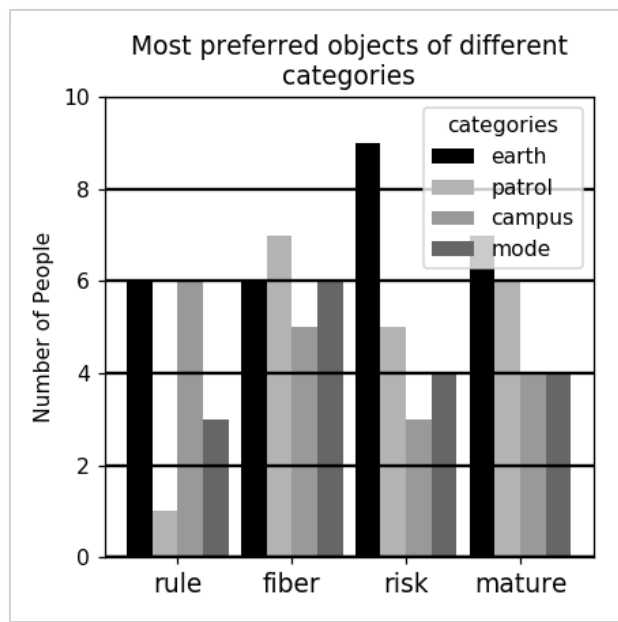
Visual Impairment



Digital Agent

Limitations of Existing Datasets

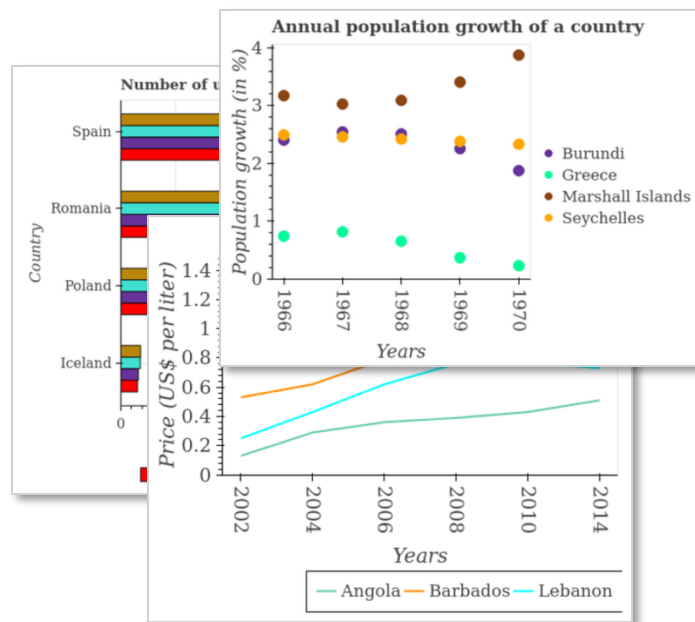
Unrealistic



(DVQA. Kahou et al. 2018)

- Random values and labels.
- Lack of semantics.

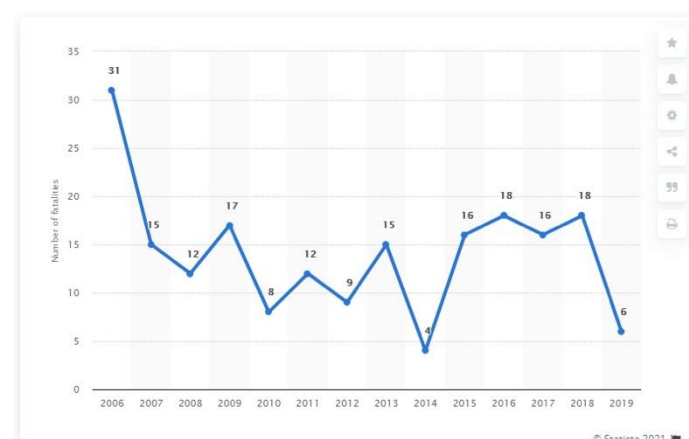
Lack of Diversity



(PlotQA. Methani et al. 2020)

- Limited types of figures.
- Template-based questions.

Limited Supervision



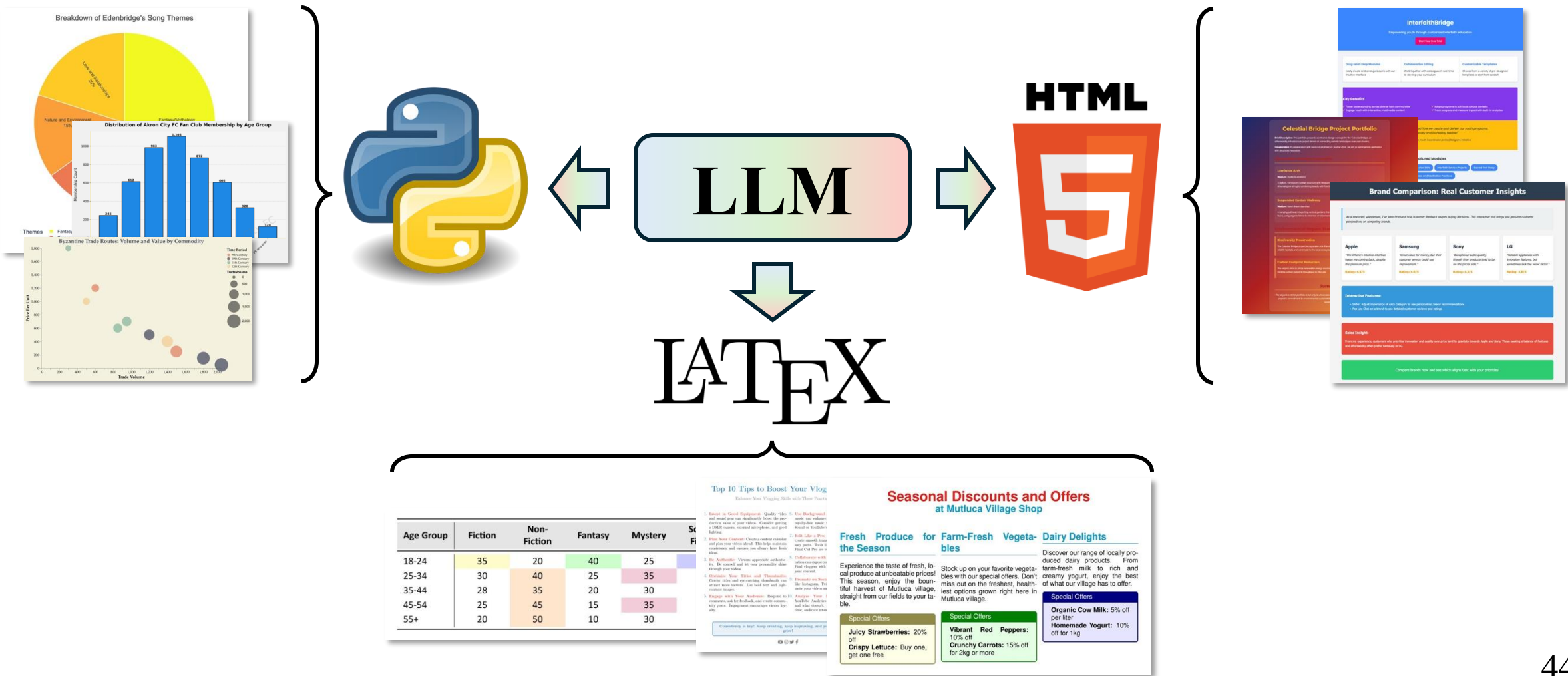
Q: What is the average of all fatalities in all years? A: 14

(ChartQA. Masry et al. 2022)

- Lack of intermediate reasoning annotations.

Motivation

Text-rich images are usually generated programmatically.



Motivation

Priors

Text Reasoning vs. Visual Reasoning

Text Input

Province	Recycling Rate (%)
Zhejiang	57.8
Jiangsu	52.3
Shandong	48.9
Guangdong	45.6
Beijing	43.2
Shanghai	41.7
Fujian	39.5
Tianjin	37.8
Hubei	35.1
Sichuan	33.4
Liaoning	31.9
Henan	29.7
Shaanxi	27.5
Yunnan	25.3
Gansu	23.1

LLM

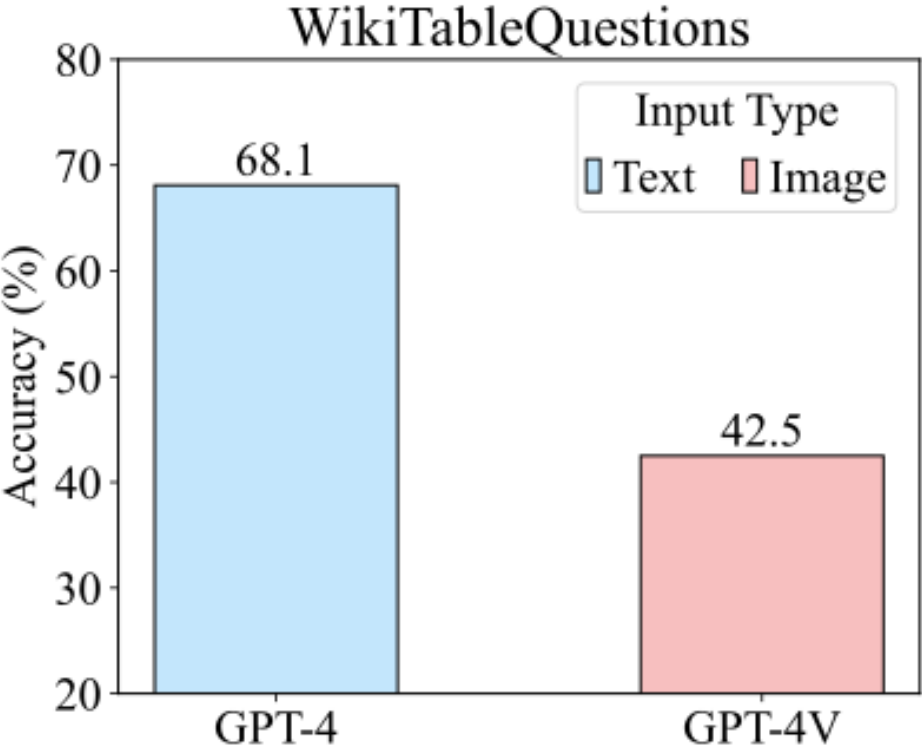
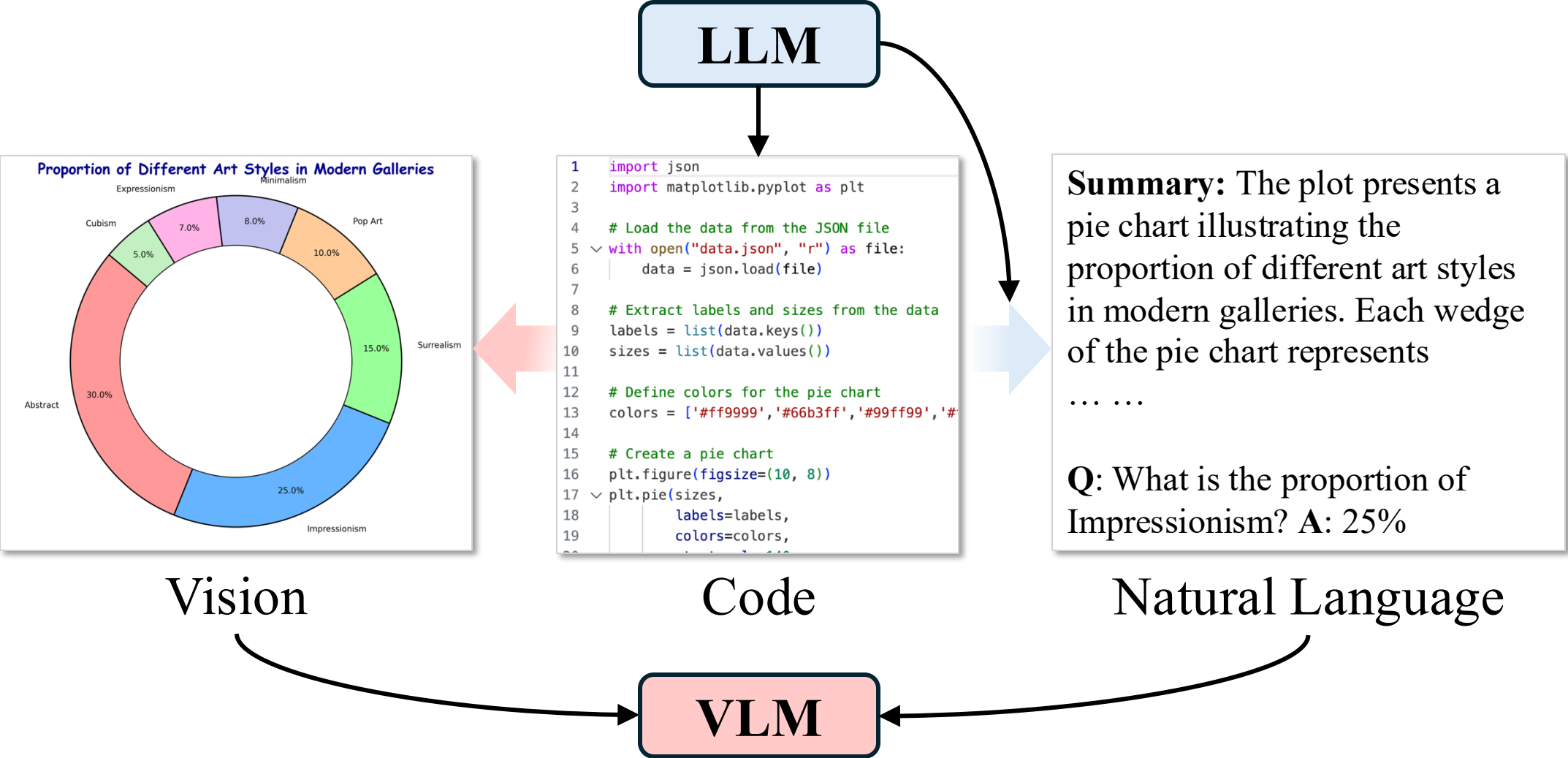


Image Input

Province	Recycling Rate (%)
Zhejiang	57.8
Jiangsu	52.3
Shandong	48.9
Guangdong	45.6
Beijing	43.2
Shanghai	41.7
Fujian	39.5
Tianjin	37.8
Hubei	35.1
Sichuan	33.4
Liaoning	31.9
Henan	29.7
Shaanxi	27.5
Yunnan	25.3
Gansu	23.1

VLM

Code as the Intermediate Representation



Code Guided Synthetic Data Generation

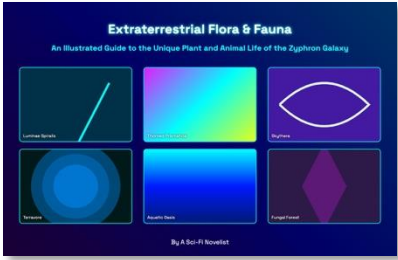
- **Input:** prompt to describe the types of images, e.g., “book cover”, “receipt”, etc.
- **Output:** A multimodal dataset with images and instruction-tuning data.



Give me a dataset of **book covers**.

Generated Topic:
A novel about “Extraterrestrial Flora & Fauna: An Illustrated Guide to the Unique Plant”.

Generated Data:
{author: sci-fi novelist, key elements: Luminae Spiralis, Skythera, ... }



Topic Generation > Data Generation > Code

Selected Persona: A sci-fi novelist constantly bouncing off ideas for new alien worlds.

```
<!DOCTYPE html> <html lang="en"> <body>
...
</body> </html>
```

Textual Instruction

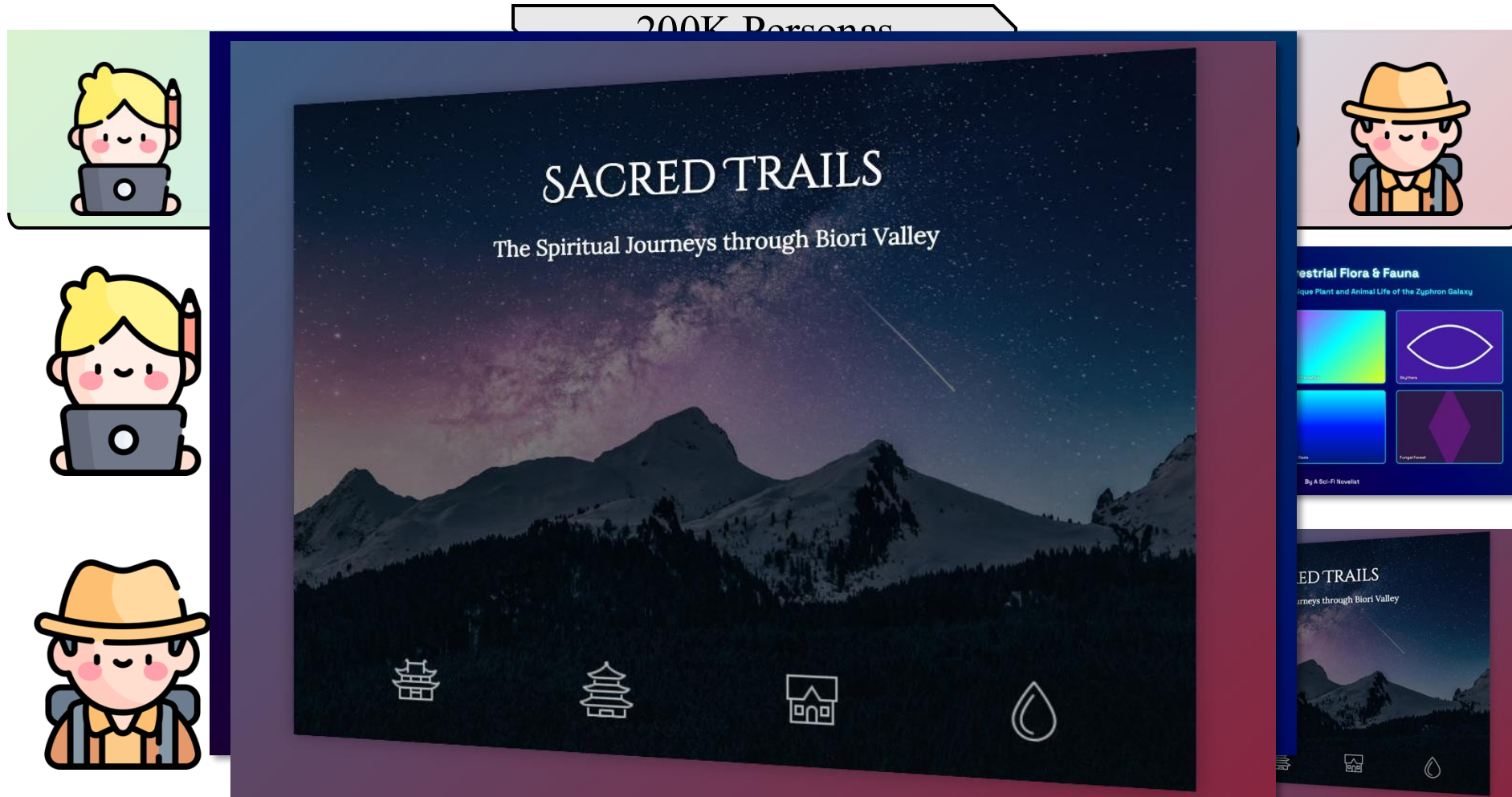
11 Tools



Q: How many illustrated elements are shown on the book cover? **A:** 6.

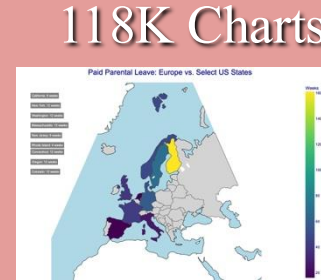
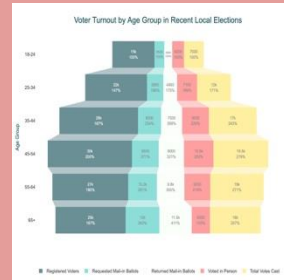
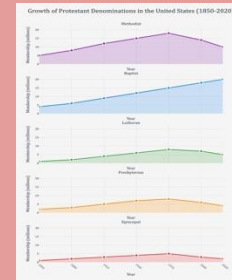
Use Personas to Enhance Diversity

Goal: Generate Diverse Images of **Book Cover**

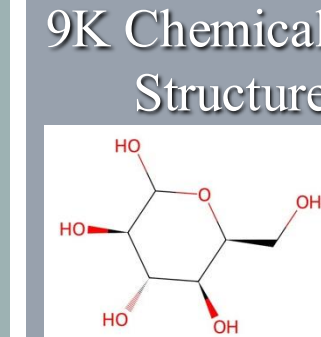
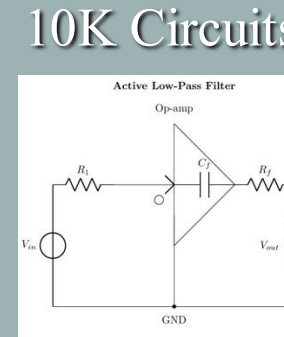
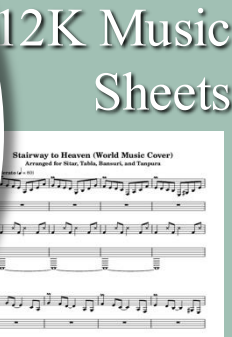
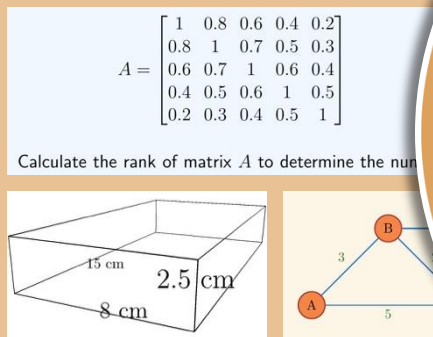
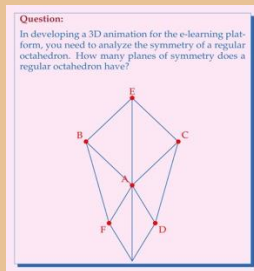
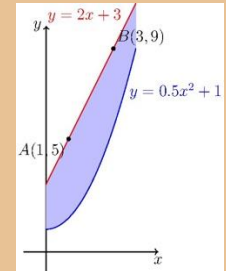


Large Scale Synthetic Dataset

73K Documents



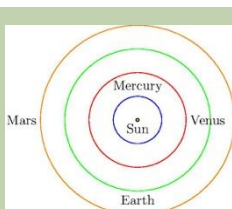
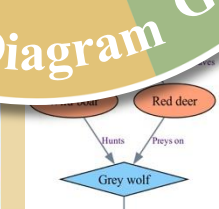
68K Math Questions



Industry	Application	Market Size (Million USD)	Growth Rate (%)	Key Players	Usage Frequency
Agriculture	Crop monitoring and health assessment	1,200	14	DJI, Precision Agriculture, John Deere	Weekly
Construction	Site surveillance and mapping	950	20	BlackVue, DroneDeploy, SenseFly	Daily
Logistics	Last-mile delivery	1,600	30	Amazon, FedEx, DHL, UPS	Daily
Insurance	Damage assessment and risk evaluation	750	18	Kellogg, Skyline, Zillow	Monthly
Public Safety	Search and rescue operations	400	15	Dragonfly, Full, Skyline	Occasional
Energy	Infrastructure inspection (e.g., power lines)	600	22	DroneDeploy, Skyline	Quarterly

Engineer	Industry	Test Level	Pass Rate (%)	Language	Key Skills
John Doe	Software Development	Senior	95	Python, JavaScript	Problem Solving, Teamwork
Jane Smith	Hardware Engineering	Mid-Level	88	C++, Verilog	Circuit Design, Debugging
Mike Johnson	Mobile App Development	Junior	72	Swift, Kotlin	UI/UX Design, API Integration
Sarah Lee	Cloud Architecture	Senior	92	Azure, AWS	Cloud Migration, Security
David Kim	Data Science	Mid-Level	85	R, Python	Statistical Analysis, Machine Learning
Emily White	Quality Assurance	Junior	78	TestNG, JUnit	Test Case Design, Bug Reporting
Chris Brown	System Administration	Senior	90	Linux, Bash	Network Troubleshooting, Server Management
Alex Green	DevOps	Mid-Level	82	Docker, Kubernetes	CI/CD Pipeline, Infrastructure as Code
Olivia Black	Product Management	Senior	98	Agile, Scrum	Market Research, User Stories
Noah Grey	Business Development	Junior	75	Sales, Negotiation	Lead Generation, Client Relations

Month	Elegance Boutique	Lavish Fashions	Sunshine Styles	Coastal Couture
January	4500	3200	5800	3900
February	4800	3000	6050	4100
March	5200	3300	6200	4300
April	5300	3400	6500	4500
May	5400	3500	6700	4600
June	5550	3600	6900	4800
July	5200	3400	6830	4500
August	5000	3300	6600	4400
September	4900	3200	6450	4300
October	5100	3350	6650	4450
November	5300	3600	6850	4600
December	5800	4000	7100	4900



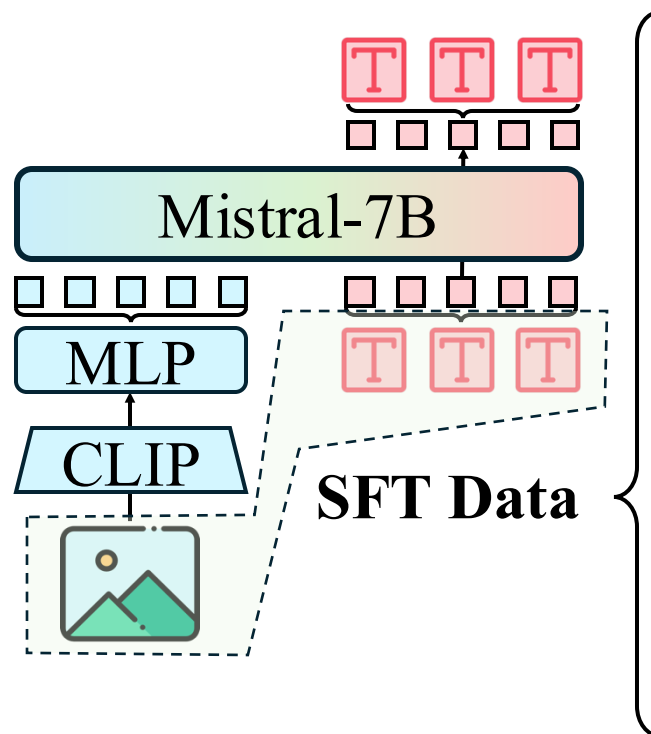
48K Tables

36K Diagrams

28K Vector Graphics

Experimental Setup

- **Architecture:**



- **Supervised Finetuning Data (SFT):**

Evaluation Datasets. (7 Text-rich benchmarks)
ChartQA, DocVQA, InfoVQA, AI2D, etc.

Auxiliary Datasets. VQAv2, GQA, OKVQA, ScienceQA, TabMWP, ST-VQA, TallyQA, etc.

Synthetic Datasets. Charts, Documents, Tables, Diagrams, Vector Graphics, Music Sheets, etc.

Zero-shot

- **Baselines:**



Open-source (7-12B)

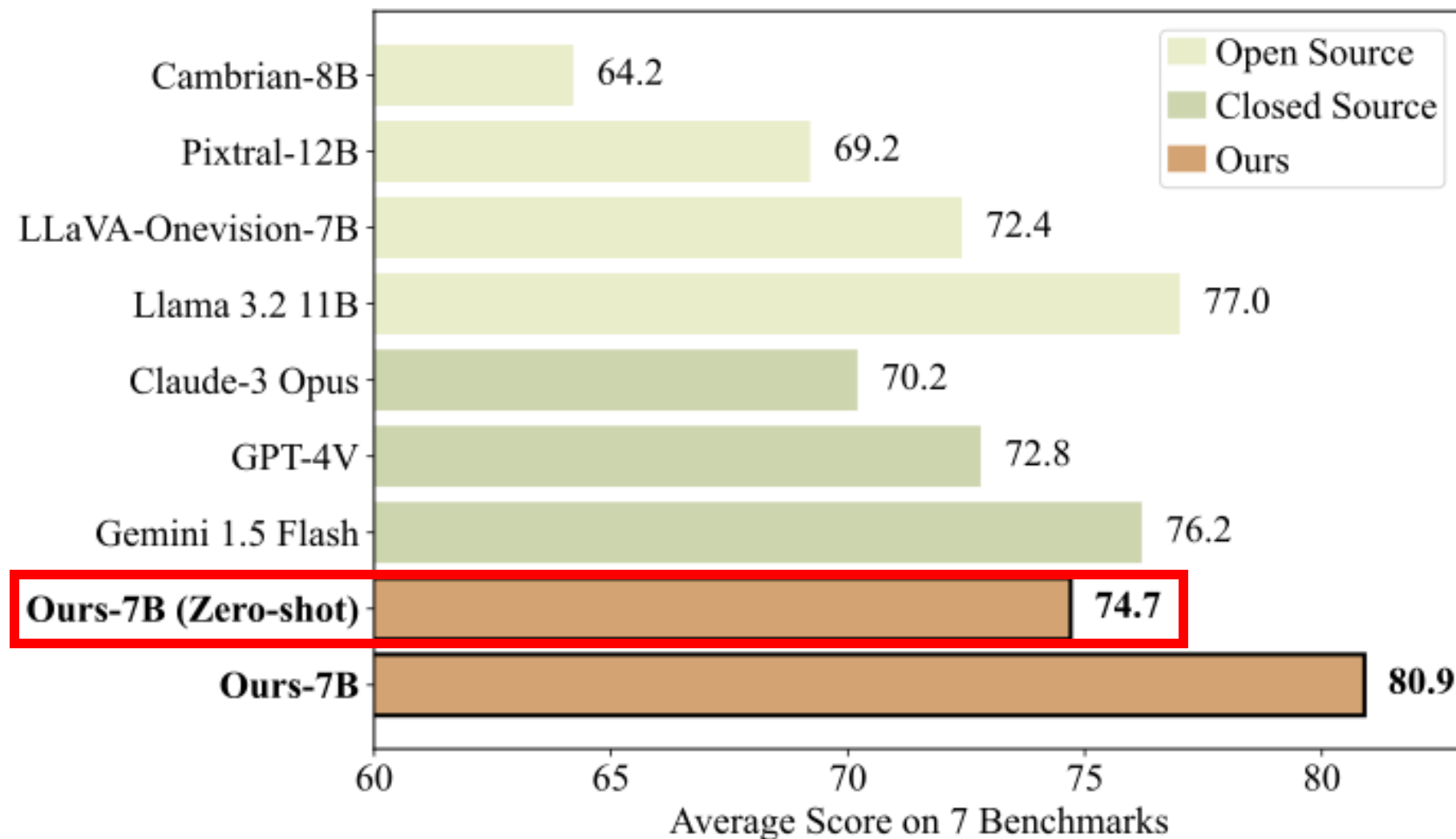
Llama-3.2-V-11B



Proprietary

GPT-4V, Gemini-1.5-Flash

Main Results



Main Results

Dataset	Ours (7B)	GPT-4V	Best Open Model
ChartQA	86.3	78.1	83.4 Llama 3.2 11B
DocVQA	90.0	87.2	90.7 Pixtral-12B
InfoVQA	70.5	75.1	68.8 LLaVA-OV-7B
TableVQA	65.8	60.5	67.0 Pixtral-12B
AI2D	91.9	89.4	91.9 Llama 3.2 11B
TextVQA	82.0	78.0	78.3 LLaVA-OV-7B
ScreenQA	80.1	41.6	87.7 Llama 3.2 11B
Average	80.9	72.8	77.0 Llama 3.2 11B

Performance Gain from Synthetic Data

SFT Data

Zero-shot

Auxiliary

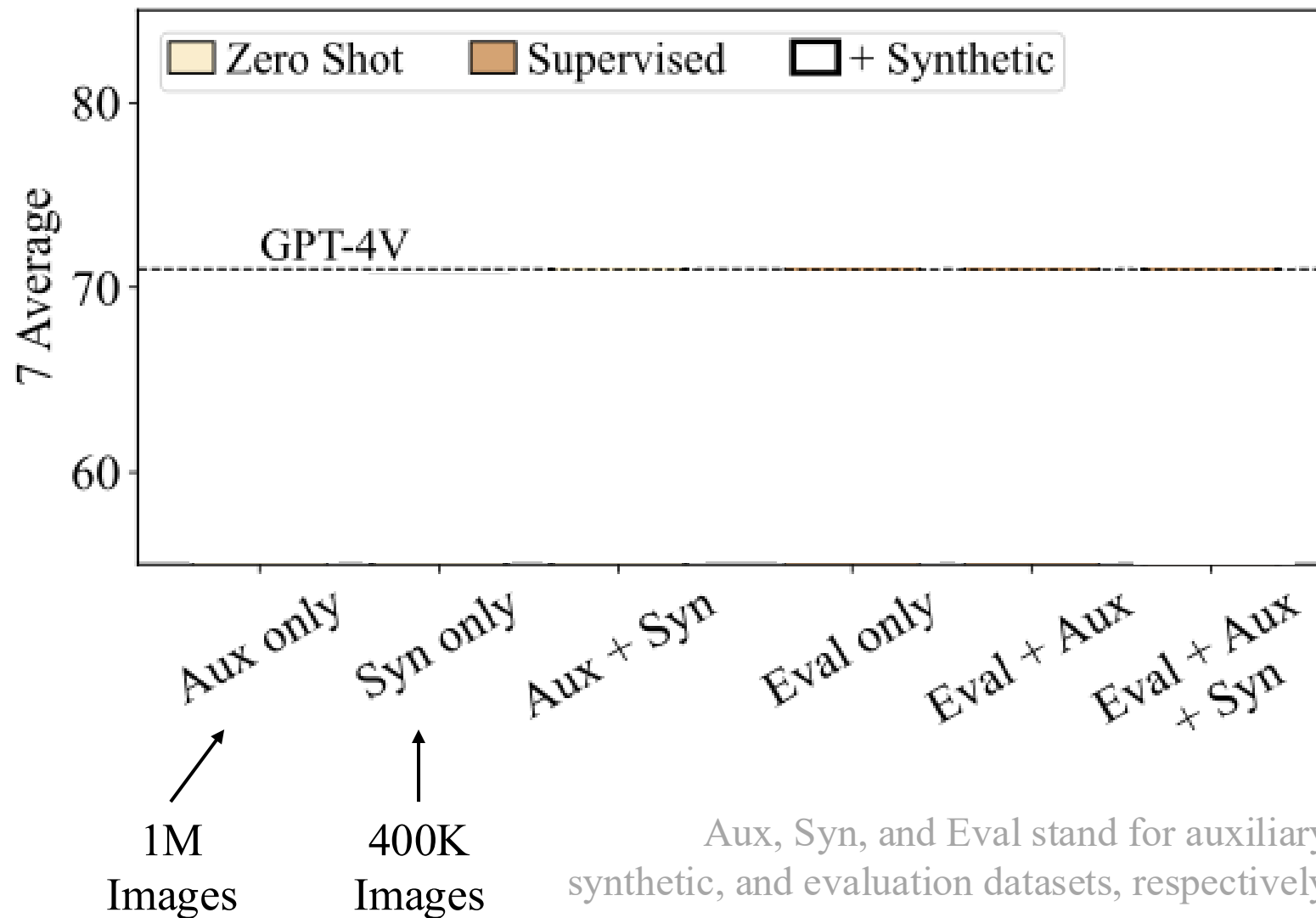
Synthetic

Supervised

Evaluation (Train)



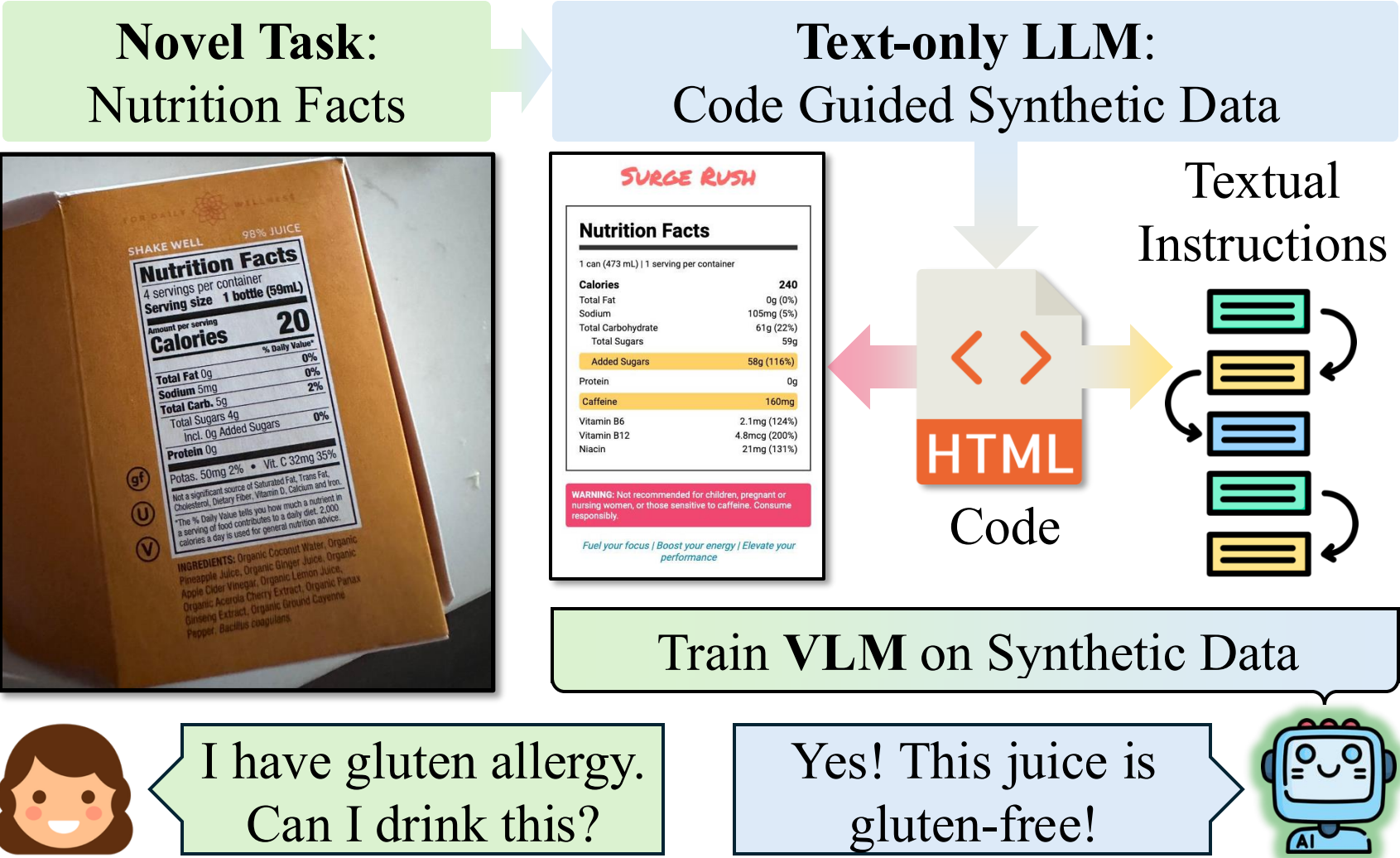
Evaluation (Test)



Performance Gain from Synthetic Data

Dataset	Academic	+ Synthetic
ChartQA	81.4	86.3
DocVQA	87.9	90.0
InfoVQA	68.2	70.5
TableVQA	53.6	65.8
AI2D	91.6	91.9
TextVQA	81.8	82.0
ScreenQA	77.0	80.1
Average	77.3	80.9

Synthetic Data to Bridge Domain Gaps



Synthetic Data to Bridge Domain Gaps

- **NutritionQA:**
 - 100 Examples, some requires multi-hop reasoning.



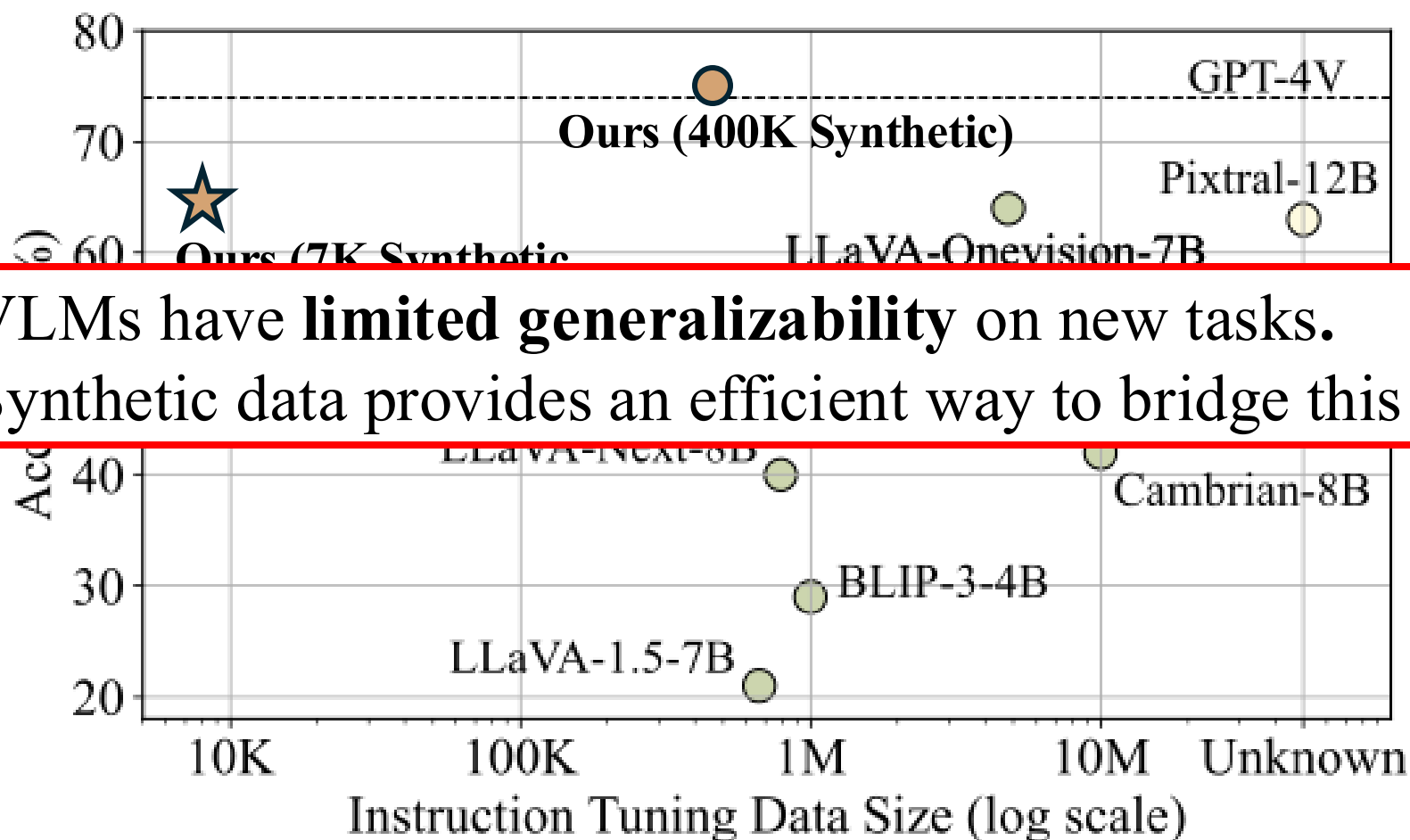
- What's the amount of iron per serving?
- How many servings do I need to fulfill the daily value of Cholesterol?



- I have taken 1000mg of sodium today. Can I eat this without exceeding the suggested daily value?

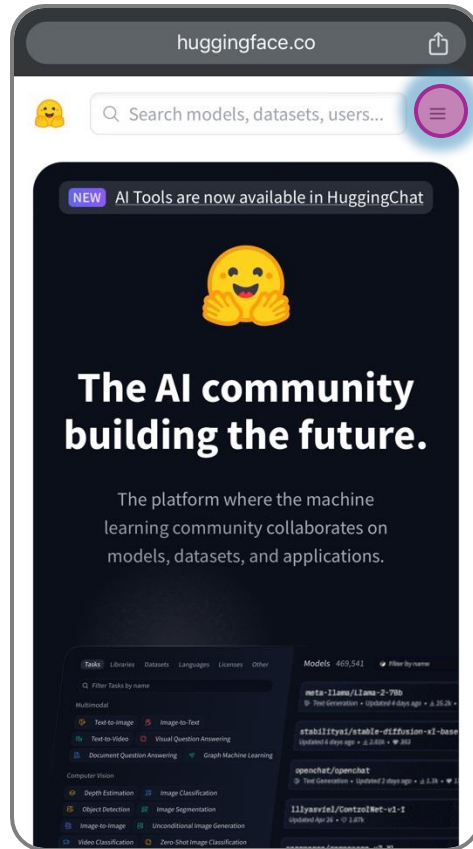
Synthetic Data to Bridge Domain Gaps

Zero-shot Performance on NutritionQA

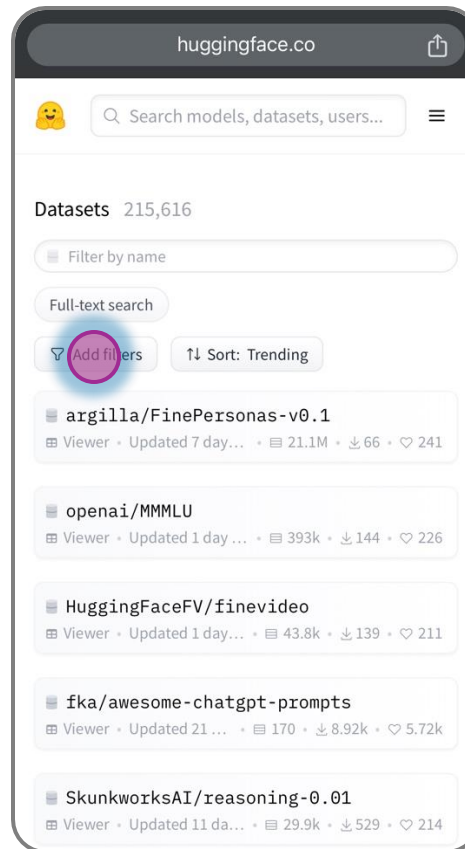


- VLMs have **limited generalizability** on new tasks.
- Synthetic data provides an efficient way to bridge this gap.

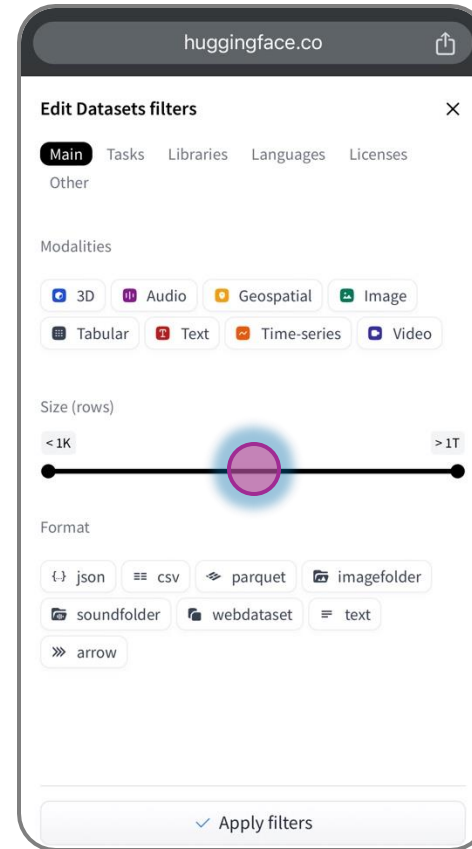
Answer with Points



“Point to the menu”

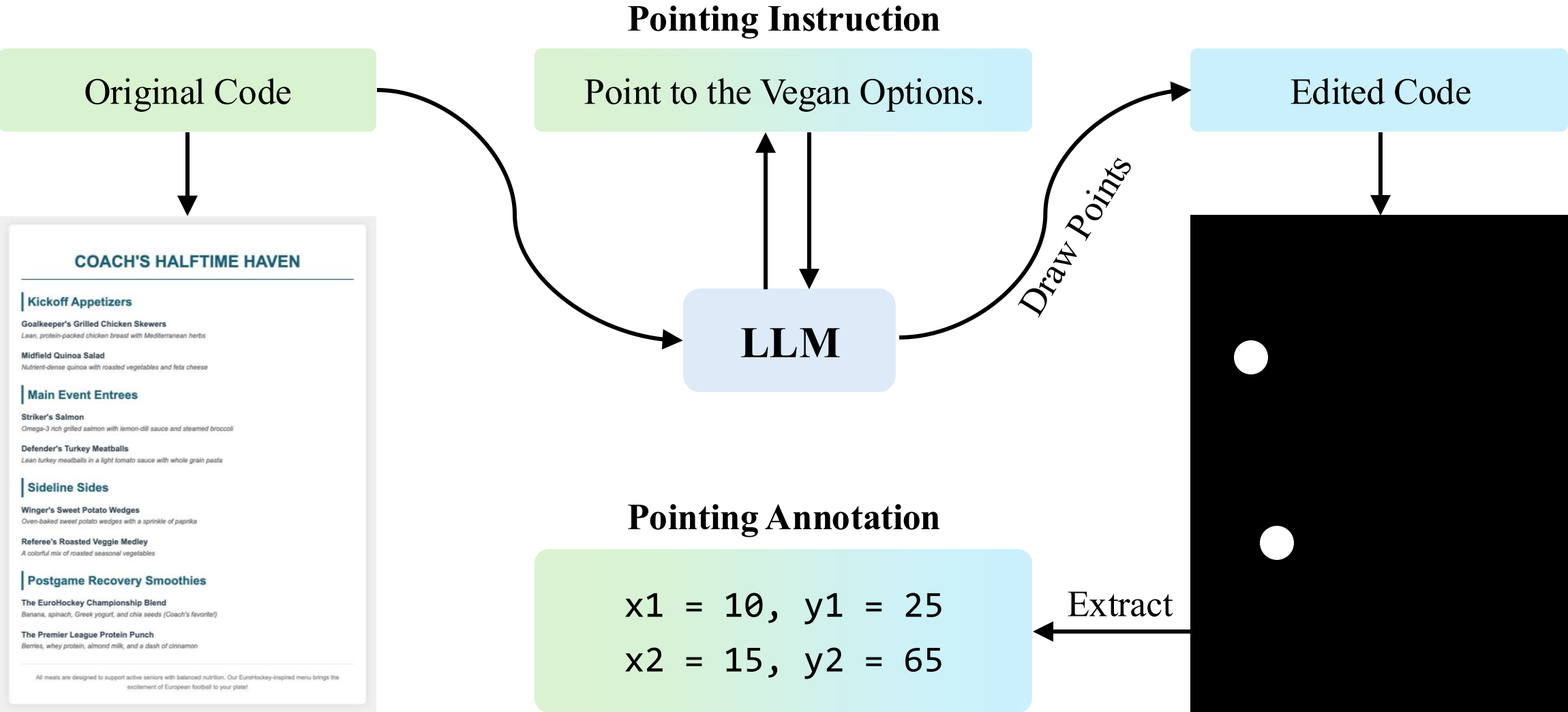


“Point to where I can set search options”



“Point to where I can find mid size datasets”

Synthetic Pointing Data

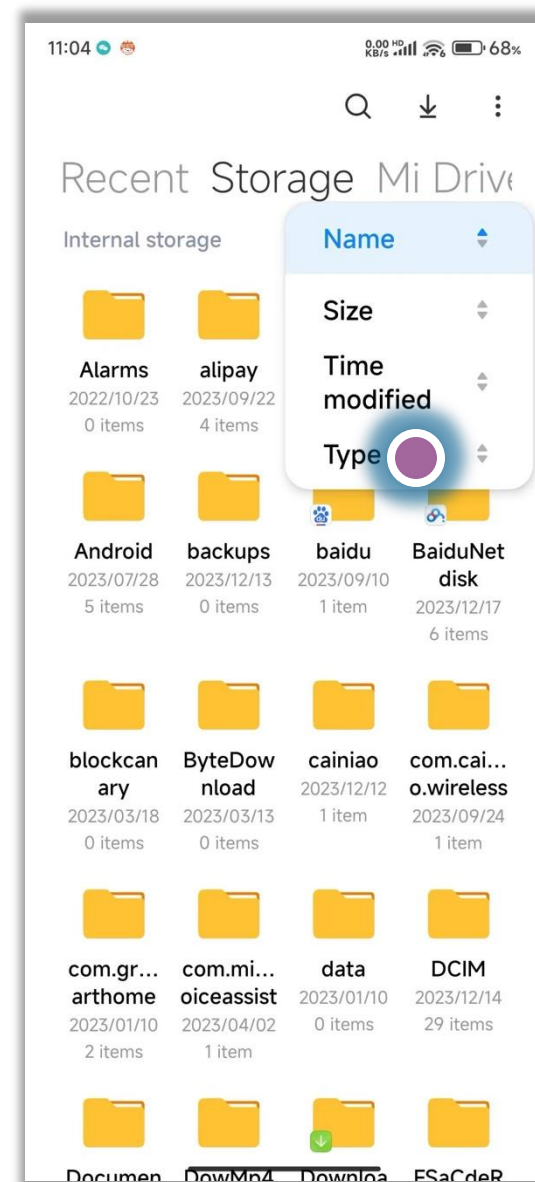


Pointing on Screen

- **Evaluation Benchmark:**
 - **ScreenSpot:** click prediction on desktop and phone.

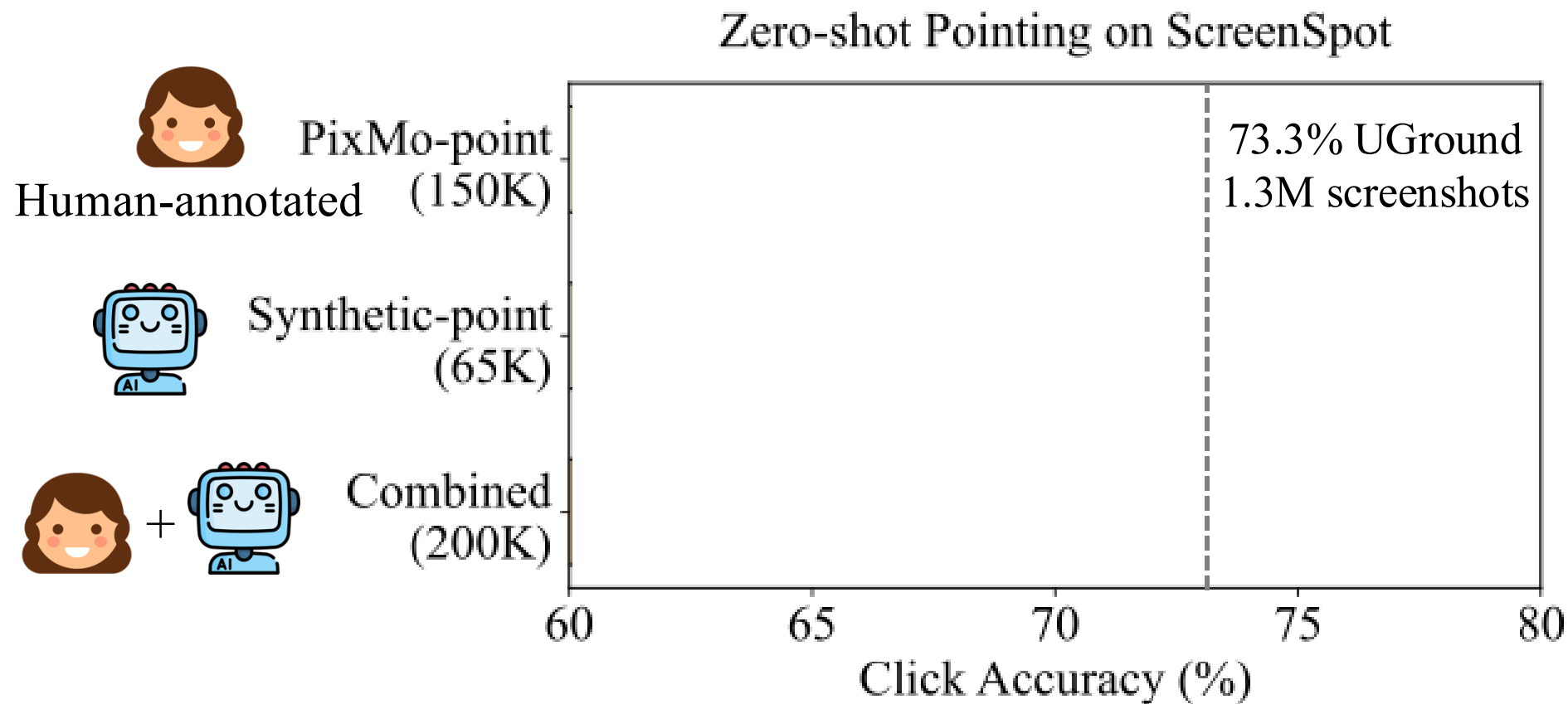


Q: Play the solitaire collection.



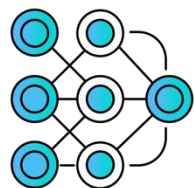
Q: Sort by file type.

Pointing on Screen



Combined means combining PixMo-point with Synthetic-point.

Talk Outline

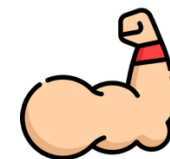


Language as
Model Structure



Interpretability

Language Bottleneck
(CVPR 23)



Robustness

Knowledge Bottleneck
(NeurIPS 24, Spotlight)



Data Efficiency

Code Guided Synthetic
Text-rich Data

VLM



Holodeck
(CVPR 24)

Embodied AI



Language to Create
Synthetic Data

Future Work

General Domain



Critical Domain

Apply **Trustworthy** Multimodal Intelligence in **Critical Domains**



Safety



Robustness



Knowledge



Interact

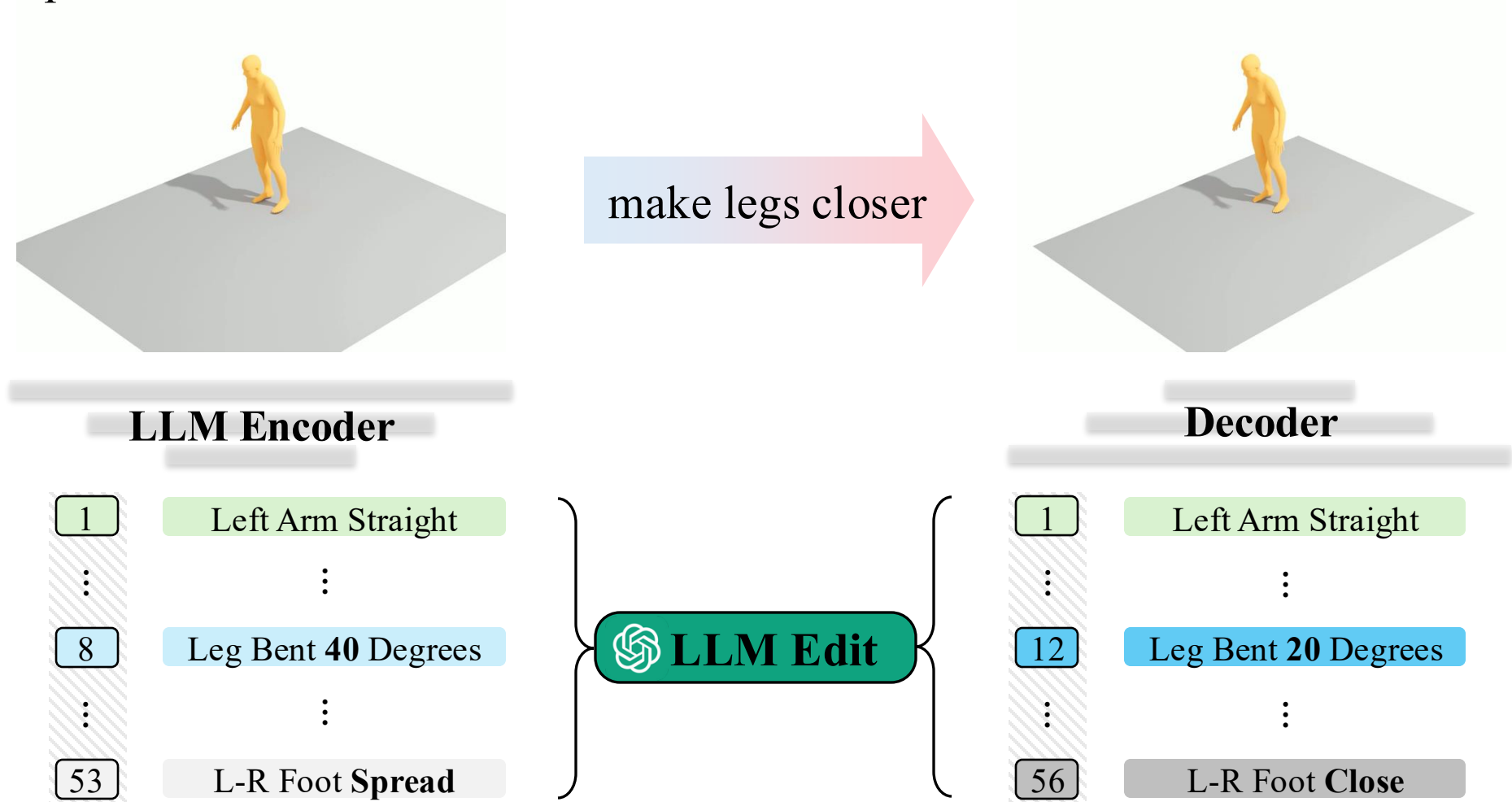
...

Controllability

Interpretable Structure

Controllable Motion Generation

A person does a handstand.



Future Work

Perceive → Reason → Act

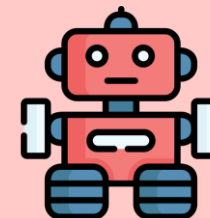
Generalizable Multimodal Intelligence that can **Interact** with Real World



Multimodal Digital Assistant

Enable multimodal models to perform diverse tasks in computer environments.

Synthetic Data

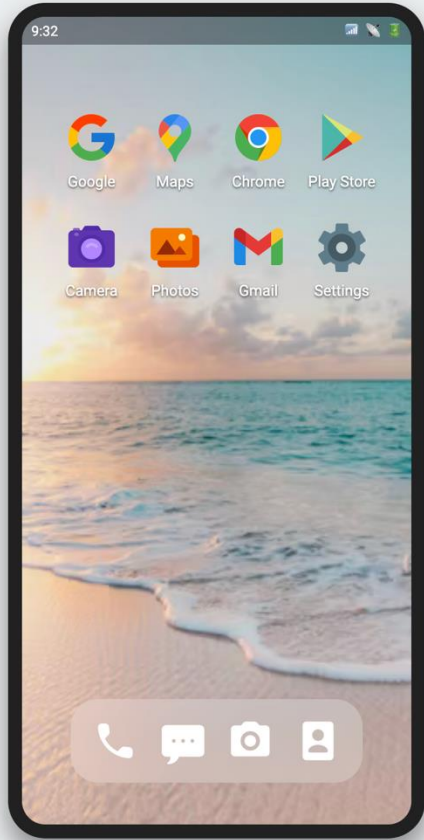


Embodied Agent

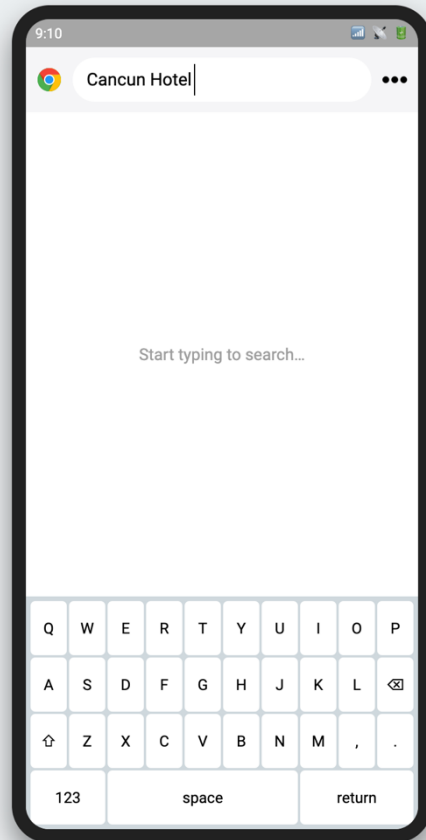
Multimodal models for agents that can navigate around and interact with the world.

Synthetic Data for UI Navigation

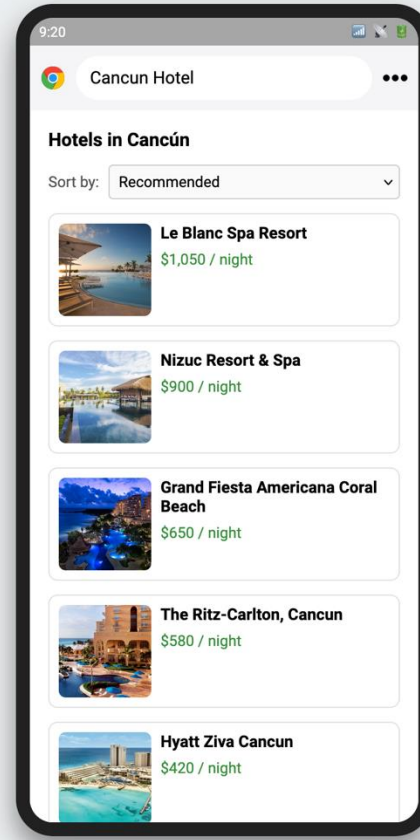
Task: Find hotels in Cancun and sort by price from low to high.



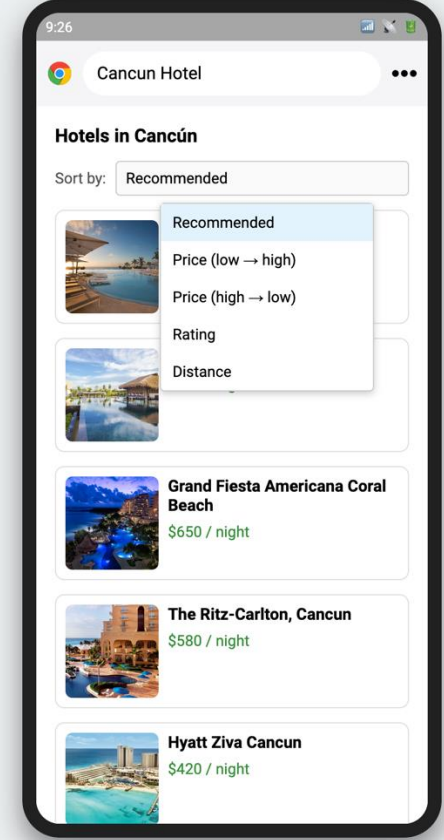
Click Chrome



Type "Cancun Hotel"



Press Enter



Change Sort by

Synthetic Data for Robot Manipulation

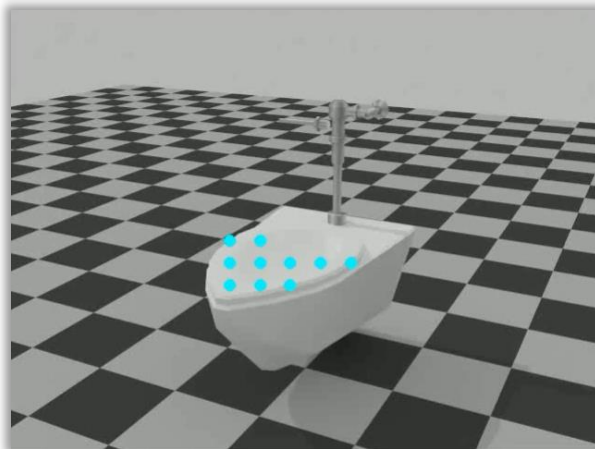
Input



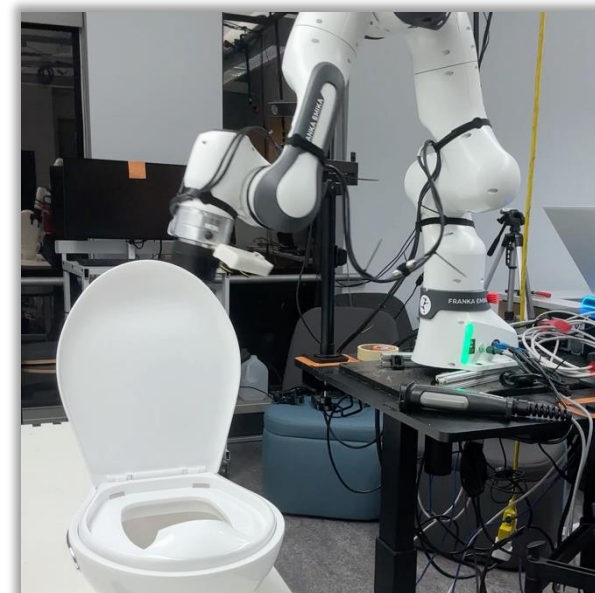
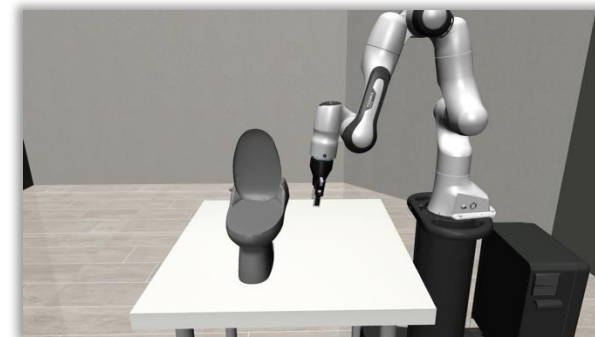
VLM

Articulated Objects

```
def partnet_102698(input_dir, links):  
    """  
    No. masked_links: 4  
    Robot Link Summary:  
    - base  
    - lever  
    - seat  
    - toilet_body  
    """  
    pred_robot = Robot(input_dir=input_dir, name="toilet")  
    pred_robot.add_link(links['base'])  
    pred_robot.add_link(links['toilet_body'])
```

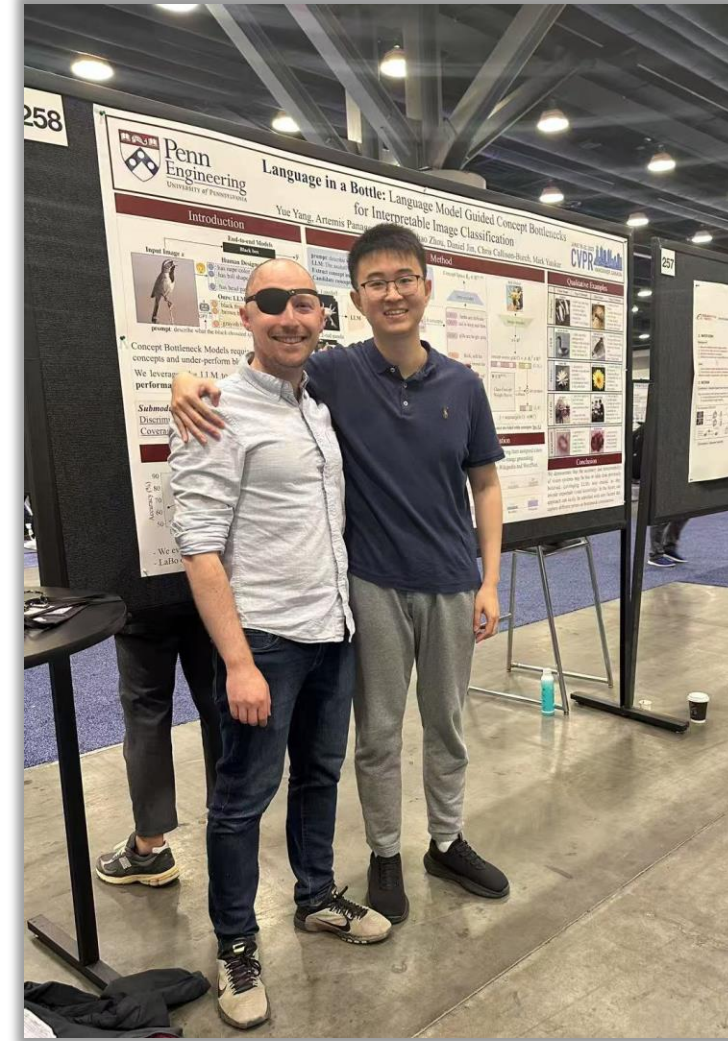
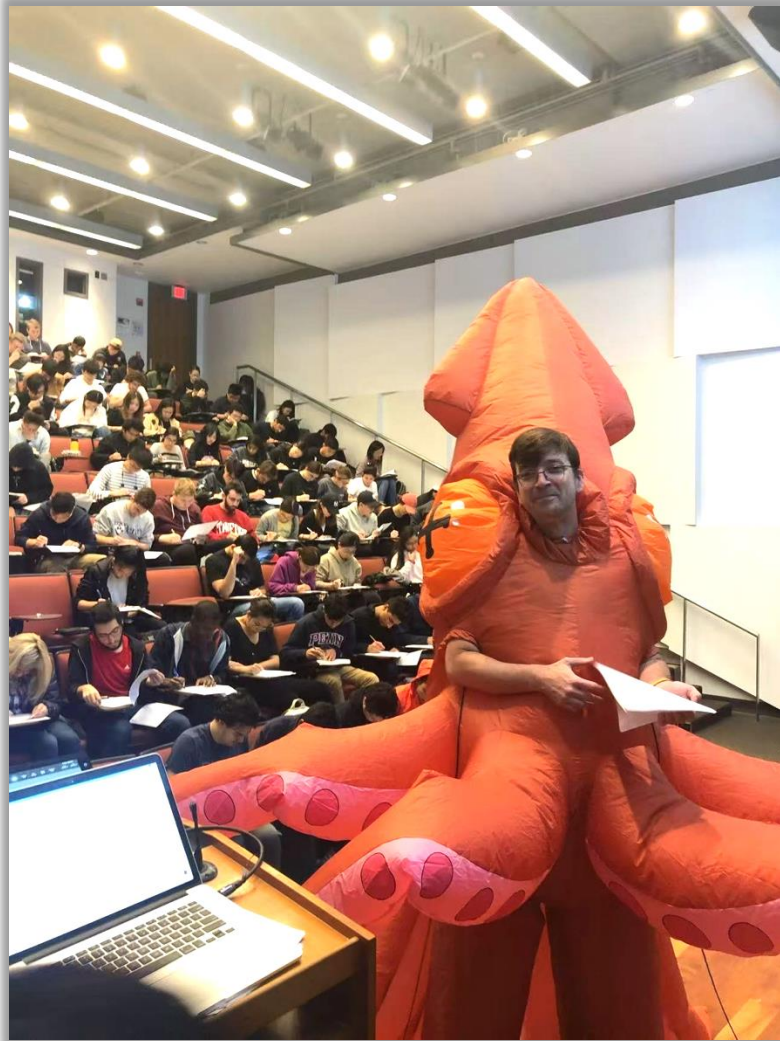


Sim-to-Real



Acknowledgement

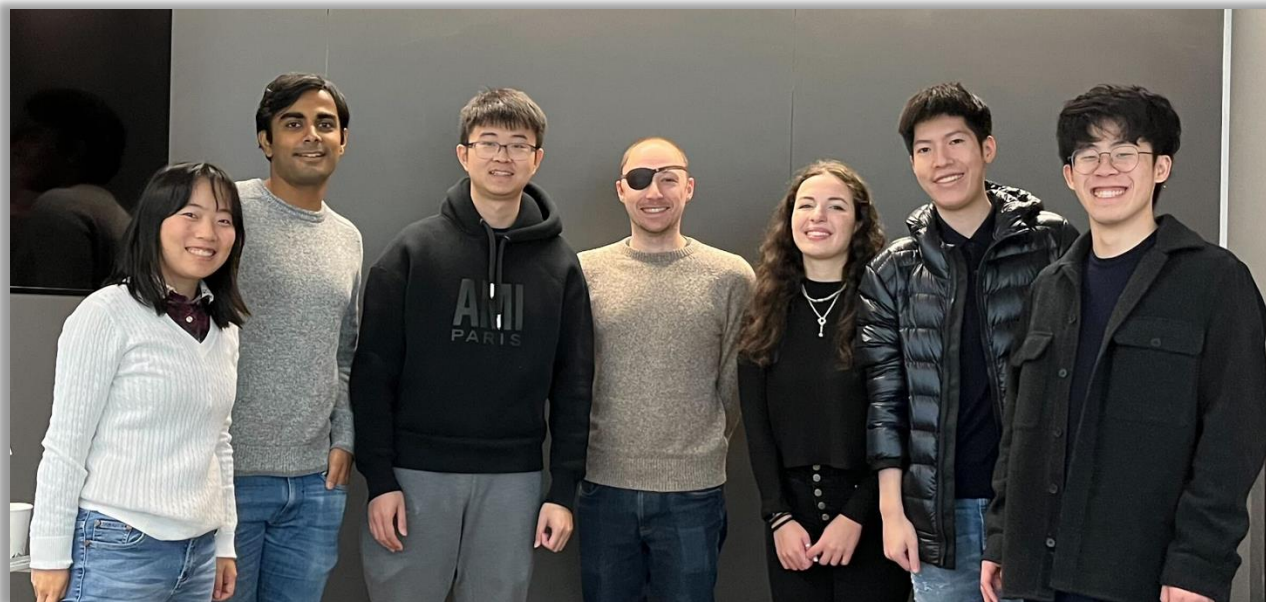
Thank you, Chris and Mark!



Thank you, committee!



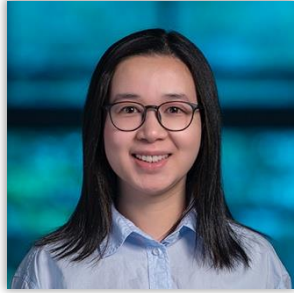
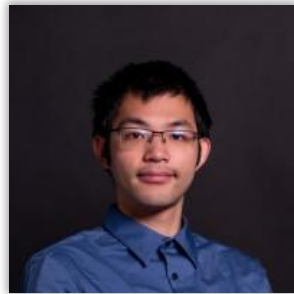
Thank you, UPenn NLP family!



Artemis, Chaitanya, Yifei,
Allen,
Josh, Weiqiu, Ajay,
Alyssa, Bryan, Liam,
Jeffery, Rebecca,
Veronica, Harry, Siyi,
Xingyu, Sihao, Ben,
Samar, Andrew, Yu,
Aditya, Delip, Marianna,
and many!

**We are not only
Research Buddies,
But also
Life Buddies!**

Thank you, collaborators!

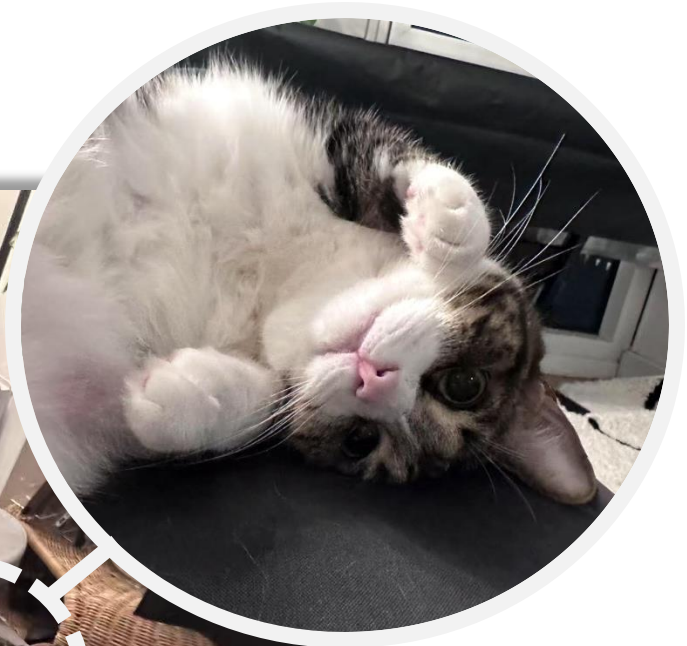


...

Thank you, my family!



Thank you, my partner!



Q & A